

Reproducibility Note for AI-Native Wireless Endogenous Security Orchestration for Low-Altitude Drone Networks

Supplementary methods and experimental records

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Abstract

This Reproducibility Note supports the architecture and case study in the main article. The opening sections explain the operational sequence, evaluation design, and principal evidence. Detailed model equations, physical assumptions, additional diagnostics, and executable records follow. Confirmation results remain separate from previously examined development data, and unfavorable outcomes are retained alongside gains. All security quantities are model outputs. The experiments do not establish secret entropy or protocol security.

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1 Reading guide and executable records

This Reproducibility Note connects the main article’s method and case study to the implemented model and archived evidence. Sections 2 and 3 provide the operational sequence and evaluation design. Sections 4 through 6 present learning, authentication, and risk diagnostics. The remaining sections specify the model, physical assumptions, additional comparisons, and reproduction records. Section numbers in this note refer to the supplement unless the main article is named explicitly.

1.1 Questions and evidence locations

The contents links open the main sections, and PDF bookmarks also expose the subsections. The following index provides a shorter route from a question in the article to the supporting evidence.

Reader question	Evidence location
How does the orchestration operate?	Section 2 explains the state, initial decisions, and executor used in main article Section III.
Which maps, policies, and costs are compared?	Section 3 describes the evaluation behind main article Section IV.
When is sensing worthwhile?	Section 4.1 gives cohort B means and paired differences supporting main article Fig. 4.
Which terms explain the utility gain?	Section 5 separates utility components and authentication phases for main article Fig. 5.
When does risk estimation become unreliable?	Section 6.1 holds observations fixed, while Section 6.2 examines online transfer and visibility stress.
What does the learning input contain?	Sections 4.2 and 15.2 give variability and the exact 37-feature specification.
What are the timing and training costs?	Sections 4.5 and 4.6 give complete-chain and offline costs. Section 20 retains separate module measurements.
Which physical assumptions remain uncalibrated?	Sections 7, 17, and 18 identify the model boundary and measurement context.
Where are the executable records?	Sections 1.3, 1.4, and 21 identify the archive and relevant files.

1.2 Version and cohort names

Cohort A is the V5 confirmation set with map seeds 107, 109, 113, 127, 131, 137, 139, and 149. Cohort B is the new V6 confirmation set with seeds 151, 157, 163, 167, 173, 179, 181, and 191. Each contains twenty missions per map at 70, 90, 110, and 130 kJ. Repeated budgets retain the same scenario identity. The earlier V4 maps support development diagnostics and are neither cohort A nor cohort B.

The original gate uses a budget portfolio for its skip branch. The revised gate uses the strongest independently trained no-sensing learner. The simple budget gate considers both threshold directions and always skipping. Validation selects always skipping. The name full denotes the residual selector after two compulsory initial sensing dwells. Fit zero means the prespecified fitting random seed zero. All new confirmation comparisons use that seed, while earlier fit variability is reported separately.

Utility is measured in objective units, not percent. Secure and exposed payload quantities are percentages of the offered 400 Mbit under the exposure model. Differences between such percentages use percentage points. Completion requires delivery and recovery within the budget. Neither modeled secure payload nor a proxy false-acceptance event establishes cryptographic confidentiality or malicious command execution. Intervals are exploratory paired bootstrap intervals without multiplicity adjustment.

1.3 Actual experiment attachment

The separate archive `WCM_ANONYMOUS_EXPERIMENTS_V6.zip` contains dependency code, frozen models, complete training and validation candidates, raw V4 through V6 evaluation records, and reproduction instructions. Its root

README provides checksum verification, one-context execution, analysis regeneration, and source-only rebuilding. The article source archive is a different attachment and does not replace these experimental dependencies.

The supplied experiment archive was extracted into a separate directory. All 1136 manifested files and 348 frozen dependency references passed hash checks. An independent execution of seven policies reproduced all eleven recorded metrics per policy for one confirmation context, giving 77 matching values. Imported project modules originated from the extracted directory. This is an executable smoke reproduction alongside complete archived results, not a claim that every training and evaluation run was repeated from scratch after extraction.

Scientific source files, candidate data, checkpoints, and frozen records retain their original bytes. Staging removes local machine paths only from nonfrozen metadata and documentation. The archive manifest records staged file hashes. Once the archive has been obtained, local reproduction does not require an external service account. The local delivery includes the archive and its SHA256 checksum file.

1.4 Current evidence files

Within the archive, current records reside under `evidence_v6_20260920/`. The following paths are relative to that directory.

- `data/feature_variation_v6.json` and `data/feature_variation_v6.csv` retain every feature range and variability classification.
- `data/learning_freeze_v6.json` and `data/confirmation_freeze_v6.json` identify model selection and the frozen confirmation configuration.
- `data/confirmation_map*_budget*.npz` retain every cohort B policy outcome. The corresponding JSON files retain phase diagnostics. `data/results_confirmation_v6.json` contains all paired groups and utility components.
- `data/auth/` retains cohort A replays under the original and continued-return-authentication rules. `data/results_auth_v6.json` gives phase and component contrasts.
- `data/robustness_statistics_v6.json`, the particle anchor and replay files, and the condition-specific raw archives retain fixed-observation and online robustness results.
- `data/learning_cost_v6_uncached_forecasts.json` gives the final complete-chain and offline cost measurements. The earlier cost file is an intermediate diagnostic with a different cache boundary and is not the source of the article's latency values.
- `data/physical_operating_region_v6.json` and its CSV companion retain the numerical reciprocity check.

1.5 Main article figure provenance

The figure provenance resides with the article source package. Main article Figs. 2 and 3 share the eligible revised-gate mission closest to the median utility of all 160 cohort B missions at 90 kJ. Eligibility requires multiple receivers and additional sensing, without filtering completion or exposure. Receiver overlays are evaluator-only, and dotted stems locate ground positions. After the initial snapshot, state tracks show post-action values, while authentication cues show pre-action values. Spatial entropy is computed from normalized occupancy probabilities and expressed in nats. Rejected-frame memory accumulates observable rejections with decay. Shading marks return after delivery, when authentication is no longer refreshed. Main article Figs. 4 and 5 read the archived paired statistics directly.

2 Operational specification of AI-native orchestration

This section expands the architecture in main article Section III into an executable decision sequence. The explanation follows observable state, initial sensing, joint configuration, and constrained execution. Section 3 then identifies how the sequence is evaluated. Detailed formulas and implementation rules are linked at the relevant steps.

2.1 Observable state and evaluator separation

Operational records contain position, altitude, remaining energy, payload progress, gateway key inventories, channel maps, receiver beliefs, and authentication records. Predictor inputs contain only the declared observable features. True receiver locations, exposed inventory fractions, and accepted forged frames are excluded from prediction inputs. Simulator outcomes nevertheless supply supervised training targets and evaluation labels. Gateway trust, bootstrap credentials, protected telemetry, and secure local inventories are deployment assumptions. Sensing associates a passive receiver with a detectable physical object. A return supplies geometric evidence without establishing malicious intent. Undetectable receivers remain a residual risk.

2.2 Initial sensing and branch selection

At launch, the revised gate constructs all 36 candidate feature vectors. The gate concatenates 19 current state entries with 12 analytic forecasts per option, giving $19 + 36 \times 12 = 451$ inputs. A random forest predicts the utility advantage of sensing over skipping. Validation selects 32 trees, minimum leaf size 16, and threshold zero. A positive prediction requests two paid sensing dwells before configuration selection. Otherwise configuration selection proceeds immediately. The dwells consume eight seconds and the modeled sensing energy.

The sensing branch uses the frozen full residual selector. The skip branch uses the independently trained no sensing residual selector. Direct utility regression is a separate comparison, not the skip branch. The original gate instead uses a budget portfolio for skipping. Gate training labels use paired branch outcomes from four whole map folds. All missions, budgets, and options of a held map remain together. Branch selectors fitted on the other maps select the held map outcomes. Validation selects gate settings before confirmation. Current confirmation uses fitting seed zero.

2.3 Joint configuration and regression targets

An option fixes a gateway, target service altitude, privacy rule, and additional sensing allowance. Two gateways and altitudes of 60, 100, and 140 m combine with six profiles.

Privacy rule	Parameter	Additional dwells
Fixed margin	0.45	0
Fixed margin	0.65	0
Fixed margin	0.80	0
Posterior quantile	0.95	0
Posterior quantile	0.95	2
Posterior quantile	0.99	4

The sensing branch ranks all 36 options. The no sensing branch ranks only the 24 options with zero additional dwells. Initial sensing is separate from the additional allowance. Configuration selection occurs once. Later feedback changes service waypoints and primitive actions without another learned option choice.

The fixed training and evaluation objective is

$$U = S + 20I - 0.7L - 5A - 0.1E. \quad (1)$$

Here S and L are modeled secure and exposed percentages of the offered 400 Mbit payload. The strict-completion indicator I equals one when delivery and recovery finish within budget and zero otherwise. The variables A and E

denote the proxy false-acceptance count and consumed energy in kJ. The fixed weights define dimensionless utility units. Incomplete missions retain the delivered secure payload contribution S , whereas the separately reported completion-adjusted metric uses SI . Direct regression predicts U . Residual regression predicts $U - V_{\text{an}}$ and adds the analytic value V_{an} back for ranking. Both frozen branch selectors use 256 trees and minimum leaf size three. Complete simulated mission outcomes provide labels without temporal difference learning.

2.4 Feature content and adaptation scope

Each option vector contains 19 current state summaries, six option descriptors, and 12 analytic forecasts. Section 15.2 gives the exact ordering and normalization, and Section 4.2 reports feature variability. Before sensing, 18 features are constant, six vary only across options, and 13 vary across contexts for a fixed option. After two dwells, nine are constant and 22 vary across contexts for a fixed option. Presensing matrices are identical for missions sharing a map and budget. Both gateway inventories are empty and the full payload remains at either selection time. Consequently, inventory and payload adaptation belongs to the executor. Authentication age and rejected frame memory are recorded but excluded from the prediction features.

2.5 Exposure and inventory interpretation

Section 10 defines the complete mapping from distance, propagation, and jitter to aggregate exposure, retained material, gateway inventories, and payload accounting. Each receiver contributes a smooth distance kernel. Aggregate exposure is one minus the product of unexposed shares. The chosen margin discards raw material before reconciliation retention. Positive true exposure above the margin, divided by the retained share, defines evaluator residual exposure. The planner instead averages residual exposure over posterior particles. Separate gateway inventories carry proportional exposure labels into transmitted payload. A 256 bit key nominally covers 4 Mbit. The construction is frequent rekeying rather than one time pad encryption. Geometric proxies do not establish secret entropy or cryptographic secrecy.

2.6 Shared execution and authentication order

Each primitive action occupies four seconds. The executor prioritizes recovery, requests authentication when freshness is at most one, schedules permitted additional sensing, and selects key generation or transmission from the inventory state. Section 14.3 specifies waypoint updates, sensing conditions, and movement order. The environment first replaces stale service requests with authentication. A separate guard then checks the action cost, horizontal recovery energy, and reserve. An unaffordable return action terminates the mission before execution. The checks do not certify collision avoidance, landing, or uncertain propulsion demand.

Freshness starts at six intervals. A challenge succeeds with probability 0.98 when the highest gateway SNR is at least 5 dB and 0.60 otherwise. Success resets freshness before forged frame evaluation. Attempts follow a Poisson process with mean 0.08 per interval. Proxy acceptance probability is 0.005 while fresh and 0.70 otherwise. Rejected frame memory uses decay 0.8 without changing acceptance probability or triggering authentication. Freshness decreases after frame evaluation. Return priority can leave authentication stale during recovery. Accepted proxy frames never execute commands. Strict completion requires payload delivery and recovery within the energy budget, independently of exposure or acceptance labels.

3 Case-study design and evidence guide

This section connects the operational sequence in Section 2 to the evaluation in main article Section IV. The sampling design distinguishes model selection, fresh confirmation, and mechanism diagnostics. The subsequent evidence sections address sensing value, utility attribution, and risk reliability using the frozen models and records.

3.1 Sampling units and evaluation roles

Training and validation use separate maps. Candidate generation evaluates two initial sensing settings and all 36 mission options. Thus $(1280 + 480) \times 2 \times 36 = 126720$ executions supply training and validation labels. Repeated budgets, options, and initial sensing settings do not create independent scenes.

Set	Maps	Scenes per map	Budgets	Contexts	Scene clusters
Training	8	40	4	1280	320
Validation	4	30	4	480	120
Cohort A	8	20	4	640	160
Cohort B	8	20	4	640	160
Inference and workload diagnostics	8	8	1	64	64
Privacy and sensing crossing	8	10	2	160	80

Training map seeds are 7, 11, 19, 47, 53, 59, 61, and 67. Validation seeds are 23, 29, 71, and 73. Section 1.2 lists the cohort A and B seeds. Cohort B was generated after model freezing. New comparisons use fitting seed zero. Map and scene identities remain paired across budgets and policies. The bootstrap uses 5000 hierarchical draws and exploratory intervals without multiplicity adjustment.

3.2 Detailed operating assumptions

Sections 8 through 13 give the city grid, flight levels, radio and channel model, key generation and exposure functions, receiver prior, sensing filter, authentication model, and operation costs. Section 18 reports the numerical reciprocity check, including the SNR requirements and modeled sampling rates. The 4 Mbit workload per fresh 256-bit key represents frequent rekeying across 500 kB segments and is an application assumption. Neither the physical rate proxy nor the observation model has been calibrated by flight measurements.

The no-sensing learner is independently trained with sensing prohibited. The budget portfolio chooses options using validation data. The original gate uses the portfolio for skipping, while the revised gate uses the no-sensing learner. Section 4 gives the new-map comparison and feature audit. Section 5 reports phase-specific authentication outcomes and the controlled return rule. Section 4.4 retains the cohort A direct-value and analytic comparisons. Section 19 retains earlier fitting variation and controlled sensing comparisons.

3.3 Workload and sensing controls

The workload and building-occlusion diagnostics in Section 6.2 use 64 earlier scenarios at 90 kJ. The gate in those diagnostics is the original gate with the portfolio skip branch. All five tested policies fail to complete under 1 Mbit per key, while all complete under 16 Mbit per key. The workload changes key demand independently of the learning architecture and must be specified before policy rankings are interpreted.

The privacy and sensing crossing in Section 19.2 uses 80 earlier scenarios at 70 and 90 kJ, with gateway B and altitude 60 m fixed. All conditions pay for two initial sensing dwells. Nine configurations cross three privacy rules with three additional sensing allowances, giving 1440 executions. At fixed 95th-percentile and 99th-percentile privacy rules, allowing up to two additional dwells improves completion-adjusted modeled secure payload by 18.11 and 24.12 percentage points, pooled across both budgets. The respective paired intervals are 6.53 to 30.00 and 10.59 to 39.51 points. The comparison isolates additional sensing and does not compare against a completely unsensed mission.

3.4 Computing and energy accounting

Sections 4.5 and 4.6 retain complete software-chain timing, model size, update cost, and candidate-generation measurements. The revised gate’s 50.03 ms result is the 95th percentile over 48 validation contexts on a desktop Intel Core Ultra 9 275HX. The measured chain includes forecasts, observation processing, option scoring, and executor initialization. Physical sensing incurs two additional 4 s dwells when selected. The full-dataset generation cost is a serial extrapolation from 576 measured candidate executions.

Sections 13 and 20 give the energy boundary. Modeled energy includes propulsion, an 18 W sensing increment, and 5 W communication or authentication increments. General computing and idle sensor electronics are excluded. Task efficiency divides total completion-adjusted modeled secure payload by total mission energy, including failed missions. Latency and compressed model size do not establish onboard deadlines or computing energy savings.

4 Learning decisions and confirmation results

This section evaluates sensing and configuration decisions on cohort B. The confirmation results are presented first, followed by the feature audit, validation choices, earlier comparisons, and computing costs. Every earlier frozen result is retained. The new gate and compact model use the original training and validation cohorts. No result from the second confirmation cohort selects a model or threshold. Every new learner uses the prescribed fitting random seed zero. The archived label fit zero refers to that model fit and never to an empty model or a zero training count. The eight new maps contain 20 missions each and four matched budgets. Thus 640 contexts represent 160 map and mission clusters. Seven methods yield 4480 policy records.

4.1 Second confirmation outcomes and uncertainty

Every interval uses 5000 paired bootstrap draws over maps and then shared mission identities. Each selected mission retains all budget conditions. The intervals are exploratory and have no multiplicity correction. Eight synthetic maps and one prescribed model fit limit generalization. An independent audit reproduces the means exactly and the interval endpoints within numerical precision.

Table 1. Second confirmation means. Payload fractions and completion are percentages. Accepted records are simulated proxy acceptances per mission.

Budget kJ	Policy	Utility	Complete	Secure	Exposed	Energy kJ	Accepted
Pooled	No sensing portfolio	96.55	96.88	94.10	4.83	85.97	0.99
Pooled	No sensing learner	98.55	96.88	94.48	5.43	85.04	0.60
Pooled	Mandatory sensing learner	93.61	88.44	91.64	3.19	77.19	1.15
Pooled	Original gate	98.83	96.25	94.79	3.66	86.35	0.80
Pooled	Revised gate	98.71	96.56	95.12	4.57	80.99	0.88
Pooled	Budget threshold gate	98.55	96.88	94.48	5.43	85.04	0.60
Pooled	Compact learner	98.74	96.88	94.65	5.26	86.51	0.59
70	No sensing portfolio	81.83	87.50	86.55	9.14	67.61	1.81
70	No sensing learner	85.83	87.50	90.66	8.96	66.92	1.88
70	Mandatory sensing learner	65.39	55.00	74.27	5.90	66.24	1.82
70	Original gate	80.62	86.25	85.64	9.02	67.68	1.84
70	Revised gate	85.83	87.50	90.66	8.96	66.92	1.88
70	Budget threshold gate	85.83	87.50	90.66	8.96	66.92	1.88
70	Compact learner	85.83	87.50	90.66	8.96	66.92	1.88
90	No sensing portfolio	91.75	100.00	93.04	6.96	72.00	1.84
90	No sensing learner	98.31	100.00	92.46	7.54	74.36	0.29
90	Mandatory sensing learner	102.06	98.75	96.73	2.42	73.43	1.07
90	Original gate	102.06	98.75	96.73	2.42	73.43	1.07
90	Revised gate	101.35	98.75	95.64	3.51	72.70	0.86
90	Budget threshold gate	98.31	100.00	92.46	7.54	74.36	0.29
90	Compact learner	98.31	100.00	92.46	7.54	74.36	0.29
110	No sensing portfolio	106.31	100.00	98.40	1.60	102.14	0.15
110	No sensing learner	104.95	100.00	97.18	2.82	97.58	0.10
110	Mandatory sensing learner	103.86	100.00	97.97	2.03	84.30	0.85
110	Original gate	106.31	100.00	98.40	1.60	102.14	0.15
110	Revised gate	104.02	100.00	97.09	2.91	92.89	0.35
110	Budget threshold gate	104.95	100.00	97.18	2.82	97.58	0.10
110	Compact learner	104.95	100.00	97.18	2.82	97.58	0.10
130	No sensing portfolio	106.31	100.00	98.40	1.60	102.14	0.15
130	No sensing learner	105.10	100.00	97.60	2.40	101.28	0.14
130	Mandatory sensing learner	103.13	100.00	97.60	2.40	84.78	0.86
130	Original gate	106.31	100.00	98.40	1.60	102.14	0.15
130	Revised gate	103.64	100.00	97.08	2.92	91.44	0.45
130	Budget threshold gate	105.10	100.00	97.60	2.40	101.28	0.14
130	Compact learner	105.88	100.00	98.29	1.71	107.17	0.10

Table 2. Paired pooled differences with 95 percent intervals. The first policy is the minuend. Completion and exposure differences are percentage points.

Comparison	Utility	Complete pp	Exposed pp	Energy kJ
No sensing learner minus No sensing portfolio	2.00 [−2.74, 6.85]	0.00 [−9.38, 9.38]	0.60 [−0.83, 2.20]	−0.94 [−2.33, 0.44]
Revised gate minus No sensing learner	0.16 [−2.27, 2.17]	−0.31 [−1.09, 0.00]	−0.86 [−2.02, 0.12]	−4.05 [−8.20, −0.76]
Revised gate minus Mandatory sensing learner	5.10 [1.63, 9.21]	8.12 [2.34, 14.22]	1.38 [0.06, 3.10]	3.80 [1.53, 6.26]
Revised gate minus Original gate	−0.12 [−3.99, 4.65]	0.31 [−9.06, 9.53]	0.91 [0.09, 1.90]	−5.36 [−8.47, −2.68]
Compact learner minus No sensing learner	0.19 [−0.15, 0.57]	0.00 [0.00, 0.00]	−0.17 [−0.45, 0.12]	1.47 [−0.40, 2.97]
Budget threshold gate minus No sensing learner	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]

Table 3. Budget specific paired differences in the original utility.

Budget kJ	Comparison	Difference and interval
70	No sensing learner minus No sensing portfolio	4.00 [−7.06, 19.97]
70	Revised gate minus No sensing learner	0.00 [0.00, 0.00]
70	Revised gate minus Mandatory sensing learner	20.44 [7.53, 36.30]
70	Revised gate minus Original gate	5.21 [−6.20, 21.03]
70	Compact learner minus No sensing learner	0.00 [0.00, 0.00]
90	No sensing learner minus No sensing portfolio	6.56 [−1.14, 14.62]
90	Revised gate minus No sensing learner	3.05 [−2.58, 9.47]
90	Revised gate minus Mandatory sensing learner	−0.71 [−4.43, 2.05]
90	Revised gate minus Original gate	−0.71 [−4.43, 2.05]
90	Compact learner minus No sensing learner	0.00 [0.00, 0.00]
110	No sensing learner minus No sensing portfolio	−1.36 [−3.34, 0.11]
110	Revised gate minus No sensing learner	−0.94 [−3.58, 0.54]
110	Revised gate minus Mandatory sensing learner	0.16 [−2.29, 2.34]
110	Revised gate minus Original gate	−2.29 [−5.40, −0.02]
110	Compact learner minus No sensing learner	0.00 [0.00, 0.00]
130	No sensing learner minus No sensing portfolio	−1.21 [−3.21, 0.23]
130	Revised gate minus No sensing learner	−1.46 [−4.20, 0.17]
130	Revised gate minus Mandatory sensing learner	0.51 [−2.14, 3.09]
130	Revised gate minus Original gate	−2.67 [−5.81, −0.27]
130	Compact learner minus No sensing learner	0.77 [−0.60, 2.27]

The no sensing learning advantage over the no sensing portfolio is not clearly reproduced on these new maps. The utility difference interval includes zero. The revised gate also has an interval crossing zero relative to the no sensing learner. The energy decrease in that comparison accompanies a slightly lower mean completion rate. These observations do not establish a universally superior gate.

The revised gate chooses initial sensing in 6.25 percent of validation contexts and 31.25 percent of second confirmation contexts. This shift reflects different map and budget contexts. The result does not identify hidden threats before observation. Relative to mandatory sensing, the gate improves mean completion while increasing modeled exposure and mission energy. These costs must accompany any completion claim.

The compact learner has the same aggregate completion rate as the larger no sensing learner. Its utility difference interval contains zero. Similar observed means do not prove formal equivalence or noninferiority. Section 4.5 reports the resource reduction separately.

4.2 Decision features and the scope of adaptation

The 37 option features contain 19 current state summaries, six option descriptors and 12 analytic forecasts. Overall variation across candidate rows can come from option identity. We therefore also fix the option and measure

variation across contexts. The state standard deviation below is the square root of the mean within option variance. It does not count changes caused solely by changing the candidate option.

Table 4. Feature variation across complete candidate matrices. The final column counts map and budget cells with any variation between missions.

Cohort and prefix	Contexts	Constant	State varying	Varying cells
Training prefix 0	1280	18	13	0
Training prefix 2	1280	9	22	32
Validation prefix 0	480	18	13	0
Validation prefix 2	480	9	22	16
Earlier confirmation prefix 0	640	18	13	0
Earlier confirmation prefix 2	640	9	22	32

At both decision times, gateway inventories are zero and the remaining payload fraction is one. The launch cell and altitude are fixed. These data do not train an inventory conditioned option selector or establish learned gains from changing remaining payload. The common executor subsequently responds to inventory and payload changes through deterministic rules.

Before sensing, all missions with the same map and budget have identical feature matrices. The no sensing learner and the initial gate can use map channel forecasts and budget. These selectors cannot distinguish hidden threats in missions with identical initial observations. The sensing branch obtains mission specific posterior evidence after two paid sensing actions. The learned mission option is selected once. Later waypoint updates belong to the shared rule based executor.

Table 5. All 37 training features. Each range gives minimum and maximum. SD fixes option identity before measuring context variation. SD values below 10^{-10} are displayed as zero. Unrounded values, full per option records and both other cohorts remain in the machine readable feature audit.

Feature	Prefix 0 range	Prefix 2 range	SD 0	SD 2
bias	1, 1	1, 1	0	0
energy_fraction	1, 1	0.979, 0.989	0	0.0037
budget_scaled	0.538, 1	0.538, 1	0.172	0.172
remaining_payload_fraction	1, 1	1, 1	0	0
elapsed_fraction	0, 0	0.01, 0.01	0	0
c_scaled	0.08, 0.08	0.08, 0.08	0	0
r_scaled	0.84, 0.84	0.84, 0.84	0	0
current_altitude_scaled	0.5, 0.5	0.5, 0.5	0	0
gateway0_inventory_scaled	0, 0	0, 0	0	0
gateway1_inventory_scaled	0, 0	0, 0	0	0
trust_scaled	1, 1	0.667, 0.667	0	0
belief_entropy_scaled	0.814, 0.814	0.177, 0.766	0	0.183
region0_mass	0.513, 0.513	0.000106, 0.736	0	0.217
region1_mass	0.481, 0.481	0.257, 1	0	0.22
estimated_count_scaled	0.667, 0.667	0.424, 1	0	0.137
belief_mean_x_scaled	0.508, 0.508	0.163, 0.65	0	0.145
belief_mean_y_scaled	0.462, 0.462	0.33, 0.754	0	0.119
belief_std_x_scaled	0.308, 0.308	0.0922, 0.411	0	0.0462
belief_std_y_scaled	0.279, 0.279	0.0275, 0.363	0	0.0461
selected_gateway	0, 1	0, 1	0	0
selected_altitude_scaled	0, 1	0, 1	0	0
fixed_privacy_flag	0, 1	0, 1	0	0
fixed_margin	0, 0.8	0, 0.8	0	0
posterior_quantile	0.95, 0.99	0.95, 0.99	0	0
extra_sense_scaled	0, 1	0, 1	0	0
predicted_energy_fraction	0.525, 3.09	0.423, 5.7	0.341	0.346
predicted_steps_fraction	0.545, 1.61	0.449, 2.89	0.116	0.135
predicted_energy_slack	-2.09, 0.475	-4.59, 0.57	0.341	0.339
mean_leak	0.0751, 0.356	7.68×10^{-5} , 0.832	0.0479	0.162
quantile_leak	0.196, 0.869	7.09×10^{-7} , 0.935	0.106	0.148

Feature	Prefix 0 range	Prefix 2 range	SD 0	SD 2
applied_margin	0.196, 0.869	7.09×10^{-7} , 0.935	0.1	0.117
exposure_proxy	0, 0.131	0, 0.695	0.00982	0.0424
predicted_completion	0, 1	0, 1	0.403	0.428
net_key_yield_scaled	0.0661, 0.336	0.033, 0.524	0.0363	0.0523
remaining_key_need_scaled	1.04, 1.07	1.04, 1.07	0.00578	0.00576
travel_steps_scaled	0.39, 0.47	0.39, 0.47	0.012	0.0102
authentication_steps_scaled	0.44, 1.3	0.36, 2.3	0.0927	0.107

4.3 Stronger skip branch and validation selection

The revised gate retains the frozen mandatory sensing learner after two sensing actions. The skip branch is the frozen no sensing learner rather than a fixed portfolio. Supervised gate targets subtract the skip branch utility from the sensing branch utility. Both branch labels use four whole map folds. All missions, budgets and options of a held map remain together. The sensing branch outcomes reuse the earlier frozen fold archive. Four matching no sensing fits produce the new skip labels.

Gate inputs use the initial 19 observable summaries and the 12 forecasts of every option. The 451 inputs contain no map identifier, hidden receiver state or future measurement. Two forests have 32 trees and minimum leaf sizes eight or 16. The candidate thresholds are zero, two, five and ten utility units. Always skipping is an additional candidate. Validation chooses leaf 16 and threshold zero.

A simple gate compares the energy budget with one boundary at 80, 100 or 120 kJ. Both inequality directions and the always sense and always skip rules are included. Validation chooses always skipping. This control is therefore exactly the no sensing learner on confirmation. It is retained as the observed result of the simple rule comparison.

Table 6. All revised gate candidates on the original validation cohort.

Trees	Minimum leaf	Threshold	Utility	Sensing %
32	8	0	99.01	37.50
32	8	2	99.74	0.00
32	8	5	99.74	0.00
32	8	10	99.74	0.00
32	16	0	100.24	6.25
32	16	2	99.74	0.00
32	16	5	99.74	0.00
32	16	10	99.74	0.00
0	None	Skip	99.74	0.00

Table 7. All simple budget rules on the original validation cohort.

Rule	Boundary kJ	Utility	Sensing %
always skip	None	99.74	0.00
always sense	None	97.00	100.00
at most	80	98.38	25.00
at most	100	98.48	50.00
at most	120	97.79	75.00
at least	80	98.36	75.00
at least	100	98.27	50.00
at least	120	98.95	25.00

The compact learner uses the same no sensing training labels, option restrictions and residual target as the frozen 256 tree learner. The same map and mission cluster bootstrap uses fit zero. Only two compact candidates are considered, both with 32 trees and minimum leaf size eight or 16. Validation chooses leaf eight. No confirmation outcome changes this selection.

Table 8. Compact learner candidates. Time includes fitting and validation prediction.

Trees	Minimum leaf	Validation utility	Measured s
32	8	99.81	0.846
32	16	99.81	0.834

4.4 Earlier confirmation remains a separate cohort

The first confirmation cohort is retained as earlier evidence. Its residual learner does not have a clear utility advantage over the direct predictor or the rolling analytic planner. Section 19 retains the four-fit comparisons on the still earlier V4 cohort. These observations are not pooled into the new confirmation cohort.

Table 9. Earlier V5 confirmation utility contrasts, retained without retuning.

Comparison	Difference and interval
full minus direct	-0.31 [-1.97, 1.18]
full minus rolling analytic	2.92 [-1.20, 6.89]
gate minus no sensing	-1.87 [-4.02, 0.25]
no sensing minus budget no sensing	3.65 [1.59, 5.74]

4.5 Measured software decision cost

The timing workload contains 48 validation contexts on four maps, four budgets and three missions per map and budget. Each method begins after resetting the same scenario. Dynamic plan and posterior risk caches are cleared before each method. Static map and candidate caches remain available. Method order is randomized within each context. Reuse inside one complete decision is permitted.

The complete software decision measurement begins at policy reset and ends when an option and its first execution action have been selected. The measurement includes gate forecasts, gate prediction, initial observation processing when chosen, branch features, candidate scoring and executor initialization. Model loading, scenario generation and subsequent mission execution are excluded. Physical sensing dwell time is also excluded. Two sensing actions still consume eight seconds of mission time. CPU and wall times are both retained in the raw records.

Measurements use one inference thread on the recorded desktop CPU. Background host load is not controlled. These results do not certify an airborne deadline. Earlier component measurements remain component measurements and should not be added to the complete decision measurement.

Table 10. Complete software decision wall time in milliseconds. The 95th percentile is a sample latency percentile.

Policy	Samples	Median ms	95th percentile ms	Maximum ms
No sensing portfolio	48	1.00	1.55	1.73
No sensing learner	48	30.22	39.85	40.53
Mandatory sensing learner	48	33.97	42.79	43.33
Original gate	48	30.34	67.77	68.67
Revised gate	48	32.65	50.03	67.57
Budget threshold gate	48	30.85	37.10	38.60
Compact learner	48	25.24	29.30	34.97

Table 11. Serialized model size and uncompressed node plus value arrays. Process memory is larger than these arrays.

Component	Trees	File MiB	Arrays MiB
Mandatory sensing learner	256	63.854	188.252
No sensing learner	256	9.827	26.954
Original gate	256	0.411	1.107
Revised gate	32	0.053	0.138
Compact learner	32	1.229	3.371

A deployed revised gate also requires both branch models. The gate forest file alone is not the controller footprint. Reducing the no sensing forest from 256 to 32 trees reduces model size substantially, but forecast construction remains in the complete decision chain. The measured decision latency therefore does not decrease in proportion to the number of trees.

The energy model excludes general computation power. These latency and memory measurements cannot establish computing energy savings. Mission energy differences describe the modeled flight and service actions only.

4.6 Offline fitting and candidate generation

The frozen V6 fitting pipeline took 23.19 seconds. This measurement includes four no sensing fold fits, two gate candidates, two compact candidates, validation prediction, archive reads and output writing. It reuses the earlier full branch fold outcomes. The V4 model selection and final fitting archive reports 199.01 seconds. The V5 direct model and gate fitting archive reports 197.27 seconds. None of these fitting times includes the original candidate simulation campaign.

The original training labels contain 92160 policy executions over both prefix settings. Validation adds 34560 executions. Exact elapsed time for that historical generation campaign was not retained. A new serial measurement repeats 576 archived validation candidate executions across 16 context and prefix pairs. This sample takes 17.62 seconds and reproduces all archived outcome metrics. Scenario initialization and file writing are outside this sampled generation time.

Scaling the measured sample to the historical record counts gives a serial estimate of 46.98 minutes for training labels and 17.62 minutes for validation labels. These are workload extrapolations from four validation maps and two extreme budgets. They are not measured historical durations, measured parallel wall times or hardware independent cost bounds.

The current cost source is the file `learning_cost_v6_uncached_forecasts.json`. A preliminary timing file retained repeated current state forecasts across methods and is preserved only as a diagnostic record. It is not the source of the reported complete decision measurements.

5 Authentication scheduling and utility attribution

This section explains the utility attribution in main article Fig. 5 by separating authentication scheduling from payload exposure and service completion. The objective and phase definitions follow the operational model in Section 2. Cohort A contains the 640 earlier budget contexts over 160 map and mission clusters. Three frozen policies are replayed under the original schedule and a shared return authentication control. The resulting 3840 executions include 1920 original policy records. Every original record reproduces all 11 archived outcome metrics with zero maximum numerical difference. The control uses the same scenario and frozen selector. It is a mechanism diagnostic on an already inspected cohort.

Cohort B is the separate new confirmation cohort. Its 640 contexts and 160 scene clusters provide the original schedule comparison for seven frozen methods. No return authentication control is evaluated on cohort B. The two cohorts are not pooled.

5.1 Phase definition and event order

A recovery interval begins when no payload remains, remaining energy reaches the common executor recovery threshold, or forced recovery is active. The threshold is horizontal Manhattan recovery energy plus 1 kJ. An interval is also labeled recovery if the hard guard overrides the requested action to return because of energy. All other executed intervals are service intervals. These labels describe the operational phase and do not identify a causal source of a failure. A terminal call that executes no action produces no attack interval.

Each four second interval first checks the requested action and the hard recovery guard. The resulting movement or service action then executes. A successful authentication challenge resets the freshness counter before the forged frame event in that interval. Attempts and proxy acceptance labels are sampled next. The freshness counter decays after that event. Reported fresh and stale counts therefore use the counter at the event, including the effect of an authentication challenge in the same interval.

Forged attempts follow the fixed Poisson rate of 0.08 per interval. The proxy acceptance probability is 0.005 while the event counter is positive and 0.70 otherwise. Freshness starts at six intervals and a successful challenge restores six intervals. Challenge success depends on the modeled SNR. A rejected frame contributes to observable memory with decay factor 0.8. That memory does not modify the acceptance probability or the stationary arrival rate.

Attempt counts and acceptance uniforms are paired by scenario and time index. Challenge randomness has a separate stream. Different durations can therefore encounter different numbers of attempts even under the same paired process. No accepted proxy frame executes a forged command. This interface is not an implemented radio fingerprint classifier or a measured authentication attack success rate.

5.2 An affordable return authentication control

The original executor gives return movement priority over authentication after delivery or when recovery energy is limiting. The control can replace a requested horizontal return or hover action with an authentication challenge when freshness is at most one interval. The drone must still be away from recovery and forced recovery must be inactive. Remaining energy must cover the challenge, horizontal recovery and the same 1 kJ reserve. The remaining decision horizon must also allow the challenge and recovery. The unchanged hard guard subsequently checks the proposed action.

The control neither chooses a new option nor bypasses the recovery guard. At tight budgets a challenge may remain unaffordable. Across the cohort A control records, selected options and every service phase summary are unchanged. Delivered payload, modeled secure and exposed payload, and strict completion are also unchanged. Return duration, authentication effort, proxy labels and total energy can change. Thus the intervention tests dependence on a particular freshness schedule rather than replacing the authentication model.

5.3 The unchanged objective and its components

The objective is $U = S + 20I - 0.7L - 5A - 0.1E$, using the notation in Section 2. The variables S and L are modeled secure and exposed percentages of the offered payload, I is the strict-completion indicator, A is the proxy false-acceptance count, and E is consumed mission energy in kJ. The five terms retain their original weights. Removing the proxy penalty below is an explanatory accounting calculation. No model is trained or selected using that alternative score.

Table 12. Mean weighted components and total utility. The exposure, acceptance and energy entries are signed penalties. Portfolio and learner both prohibit sensing.

Cohort and schedule	Policy	S	$20I$	$-0.7L$	$-5A$	$-0.1E$	U
A original	Portfolio	94.94	19.97	-3.54	-4.30	-8.64	98.42
A original	Learner	95.83	19.72	-2.84	-2.27	-8.36	102.07
A original	Original gate	95.27	19.16	-2.45	-3.10	-8.66	100.21
A return auth	Portfolio	94.94	19.97	-3.54	-1.80	-8.83	100.74
A return auth	Learner	95.83	19.72	-2.84	-1.59	-8.42	102.71
A return auth	Original gate	95.27	19.16	-2.45	-1.82	-8.76	101.39
B original	Portfolio	94.10	19.38	-3.38	-4.95	-8.60	96.55
B original	Learner	94.48	19.38	-3.80	-3.00	-8.50	98.55
B original	Mandatory sensing	91.64	17.69	-2.23	-5.77	-7.72	93.61
B original	Original gate	94.79	19.25	-2.56	-4.02	-8.63	98.83
B original	Revised gate	95.12	19.31	-3.20	-4.42	-8.10	98.71
B original	Budget threshold	94.48	19.38	-3.80	-3.00	-8.50	98.55
B original	Compact learner	94.65	19.38	-3.68	-2.95	-8.65	98.74

Table 13. Paired component differences for the no sensing learner minus the no sensing portfolio. The change in gap subtracts the original difference from the return authentication difference.

Comparison	Weighted component	Difference and 95 percent interval
A original	Modeled secure reward	0.890 [-0.080, 1.933]
A original	Completion reward	-0.250 [-0.750, 0.000]
A original	Exposure penalty	0.702 [0.017, 1.453]
A original	Proxy acceptance penalty	2.031 [1.281, 2.773]
A original	Energy penalty	0.281 [-0.141, 0.790]
A original	Total utility	3.654 [1.586, 5.744]
A original	Utility without proxy penalty	1.623 [-0.161, 3.488]
A return auth	Modeled secure reward	0.890 [-0.080, 1.933]
A return auth	Completion reward	-0.250 [-0.750, 0.000]
A return auth	Exposure penalty	0.702 [0.017, 1.453]
A return auth	Proxy acceptance penalty	0.211 [-0.141, 0.797]
A return auth	Energy penalty	0.413 [-0.019, 0.950]
A return auth	Total utility	1.966 [0.135, 3.882]
A return auth	Utility without proxy penalty	1.755 [-0.040, 3.628]
B original	Modeled secure reward	0.380 [-2.120, 3.439]
B original	Completion reward	0.000 [-1.875, 1.875]
B original	Exposure penalty	-0.422 [-1.537, 0.579]
B original	Proxy acceptance penalty	1.945 [1.195, 2.602]
B original	Energy penalty	0.094 [-0.044, 0.233]
B original	Total utility	1.997 [-2.736, 6.852]
B original	Utility without proxy penalty	0.052 [-4.791, 5.454]
A change in gap	Modeled secure reward	0.000 [0.000, 0.000]
A change in gap	Completion reward	0.000 [0.000, 0.000]
A change in gap	Exposure penalty	0.000 [0.000, 0.000]
A change in gap	Proxy acceptance penalty	-1.820 [-2.406, -1.172]
A change in gap	Energy penalty	0.133 [0.089, 0.167]
A change in gap	Total utility	-1.688 [-2.247, -1.072]
A change in gap	Utility without proxy penalty	0.133 [0.089, 0.167]

In cohort A, the original utility gain is 3.654 units, of which 2.031 units arise from the proxy acceptance term. The sum of the other terms is 1.623 units and its interval includes zero. Under the return authentication control,

the gain is 1.966 units. The change in the gap is negative 1.688 units with a wholly negative paired interval. The proxy contribution falls to 0.211 units while the energy contribution changes because each policy incurs different extra return challenges.

In cohort B, the original utility difference is 1.997 units and its interval includes zero. The proxy term contributes 1.945 units. Neither the earlier significance nor a general payload secrecy advantage is established by the new cohort.

Table 14. Phase outcomes. Attempts, accepted labels, duration and challenges are means per mission. Acceptance percentage divides total accepted labels by total attempts. Stale percentage divides total stale event intervals by total phase intervals.

Schedule	Policy	Phase	Attempts	Accepted	Accept %	Seconds	Stale %	AUTH
A original	Portfolio	service	9.214	0.025	0.27	462.14	0.02	22.725
A original	Portfolio	recovery	1.330	0.836	62.87	71.62	89.52	0.000
A original	Learner	service	9.561	0.027	0.28	483.03	0.02	23.834
A original	Learner	recovery	0.719	0.428	59.57	38.25	85.26	0.000
A original	Original gate	service	9.639	0.027	0.28	484.80	0.02	23.859
A original	Original gate	recovery	0.973	0.594	61.00	51.64	87.15	0.000
A return auth	Portfolio	service	9.214	0.025	0.27	462.14	0.02	22.725
A return auth	Portfolio	recovery	1.573	0.334	21.25	82.44	32.31	2.705
A return auth	Learner	service	9.561	0.027	0.28	483.03	0.02	23.834
A return auth	Learner	recovery	0.794	0.291	36.61	41.43	56.52	0.795
A return auth	Original gate	service	9.639	0.027	0.28	484.80	0.02	23.859
A return auth	Original gate	recovery	1.091	0.338	30.95	57.44	45.60	1.450
B original	Portfolio	service	9.180	0.031	0.34	457.04	0.00	22.511
B original	Portfolio	recovery	1.531	0.958	62.55	74.00	87.36	0.000
B original	Learner	service	9.692	0.030	0.31	482.49	0.00	23.778
B original	Learner	recovery	0.975	0.570	58.49	47.75	80.73	0.000
B original	Mandatory sensing	service	7.959	0.030	0.37	400.15	0.00	19.583
B original	Mandatory sensing	recovery	1.723	1.123	65.19	83.07	89.22	0.000
B original	Original gate	service	9.502	0.033	0.35	473.37	0.00	23.341
B original	Original gate	recovery	1.280	0.770	60.20	61.36	84.90	0.000
B original	Revised gate	service	8.830	0.030	0.34	439.24	0.00	21.603
B original	Revised gate	recovery	1.384	0.855	61.74	67.70	86.11	0.000
B original	Budget threshold	service	9.692	0.030	0.31	482.49	0.00	23.778
B original	Budget threshold	recovery	0.975	0.570	58.49	47.75	80.73	0.000
B original	Compact learner	service	9.878	0.033	0.33	491.90	0.00	24.288
B original	Compact learner	recovery	0.972	0.558	57.40	47.12	79.95	0.000

The acceptance percentage is a ratio of pooled counts. It is not the average of mission acceptance ratios. The stale percentage uses the trust state at the forged frame event. In particular, a challenge that succeeds during an interval can make that event fresh even when the state before the action was stale. Complete per phase count, rate and duration intervals are retained in the accompanying data.

Under the original cohort A schedule, the learner has 33.375 fewer recovery seconds and 0.408 fewer recovery acceptances per mission than the portfolio. Service acceptances instead increase by 0.0016 per mission. The observed aggregate acceptance reduction is therefore concentrated in recovery. This finding explains a preference under the declared timing and scoring model rather than improved service identity classification.

Table 15. Paired phase differences on cohort A. The learner gap subtracts the portfolio. The gate gap subtracts the learner. Accepted labels and stale intervals are counts per mission. Duration differences are seconds.

Comparison	Phase	Accepted difference	Duration difference	Stale difference
Original learner gap	service	0.002 [0.000, 0.008]	20.89 [−0.12, 39.36]	0.002 [0.000, 0.008]
Original learner gap	recovery	−0.408 [−0.556, −0.258]	−33.38 [−43.87, −21.75]	−7.877 [−10.312, −5.188]
Controlled learner gap	service	0.002 [0.000, 0.008]	20.89 [−0.12, 39.36]	0.002 [0.000, 0.008]
Controlled learner gap	recovery	−0.044 [−0.161, 0.027]	−41.01 [−52.56, −27.29]	−0.805 [−2.831, 0.408]
Original gate gap	service	0.000 [0.000, 0.000]	1.77 [−20.30, 27.37]	0.000 [0.000, 0.000]
Original gate gap	recovery	0.166 [0.037, 0.295]	13.39 [2.93, 22.81]	3.098 [0.567, 5.408]

Table 16. The return authentication intervention minus the original schedule within each policy. Every service phase difference is zero.

Policy	Utility difference	Recovery acceptances	Recovery seconds	Extra challenges
Portfolio	2.320 [1.800, 2.871]	-0.502 [-0.613, -0.395]	10.82 [9.52, 11.99]	2.705 [2.380, 2.998]
Learner	0.632 [0.243, 1.107]	-0.138 [-0.241, -0.055]	3.18 [1.39, 5.64]	0.795 [0.348, 1.411]
Original gate	1.181 [0.738, 1.647]	-0.256 [-0.353, -0.164]	5.80 [4.21, 7.39]	1.450 [1.052, 1.848]

5.4 The original gate against the stronger alternative

The original gate compares sensing with the no sensing budget portfolio. Its relevant stronger external reference is the independently trained no sensing learner. The following differences use the original cohort A schedule. A lower exposed fraction accompanies lower completion and more energy, and the utility interval crosses zero. The gate also has more proxy acceptances and a longer recovery phase. These observations motivated a stronger skip branch without guaranteeing that the revised gate would dominate that branch.

Table 17. Original gate minus no sensing learner tradeoff on cohort A. All intervals are paired and use the same contexts.

Outcome	Difference and 95 percent interval
Utility	-1.868 [-4.023, 0.254]
Modeled secure payload pp	-0.564 [-1.852, 0.542]
Modeled exposed payload pp	-0.550 [-1.796, 0.575]
Strict completion pp	-2.812 [-8.594, 2.344]
Mission energy kJ	2.980 [-1.504, 8.506]
Proxy accepted labels per mission	0.166 [0.037, 0.295]
Delivered payload pp	-1.114 [-2.782, 0.046]
Mission duration s	15.162 [-8.494, 44.207]
Sensing actions per mission	1.034 [0.794, 1.309]
Authentication actions per mission	0.025 [-1.091, 1.313]
Vertical actions per mission	0.011 [-0.223, 0.284]
Completion adjusted modeled secure payload pp	-2.034 [-6.722, 2.071]

All reported intervals use 5000 paired draws that first resample maps and then complete mission identities. The same mission indices are retained across budgets and policies. The intervals are exploratory with no correction for multiple comparisons. The authentication control is not a new confirmation sample. It reuses cohort A and does not add independent scenes.

The raw archives retain attempted and accepted counts, executed action, freshness state and phase label for each executed interval of the cohort A intervention. The source summaries are results_auth_v6.json and results_confirmation_v6.json. The additional cohort B phase summaries and replay audit are in auth_supplement_derived_v6.json. None of these diagnostics establishes cryptographic command integrity or a measured physical layer authentication guarantee.

6 Inference and workload robustness diagnostics

This section separates numerical posterior approximation, inference mismatch, and changes to operating conditions. Fixed-observation replay tests risk estimates independently of policy-induced trajectory changes. Online diagnostics then test mission outcomes. The gate in this section is the original V5 gate with the portfolio skip branch. These diagnostics add the strongest no-sensing learner to earlier conditions. They do not evaluate the revised V6 gate under every shift. Fixed-observation replay, online inference mismatch, and changes to the physical sensor are separate experiments.

6.1 Fixed-observation inference diagnostics and conservative margins

These diagnostics reuse 64 previously examined map and mission pairs at 90 kJ and do not constitute fresh confirmation. A nominal full-policy trajectory fixes every observation and harvest query. Each posterior is replayed with 256 or 1024 particles and four inference seeds. Seeds are averaged within each mission before 5000 paired map and mission bootstrap draws. Harvest visits are averaged within each mission. The intervals below are unadjusted 95% intervals. Coverage is reported as a percentage. Mean leakage bias is the posterior mean minus the true modeled leakage, expressed as a dimensionless fraction. A bias of -0.09 therefore denotes underestimation by nine percentage points. The four inference seeds are 87013, 87014, 87015 and 87016. The diagnostic maps are 31, 37, 41, 79, 83, 89, 97 and 101. Each map uses mission identifiers zero through seven. Validation uses maps 23, 29, 71 and 73 with the same eight mission identifiers. The nominal sensor generator and recorded measurement sets remain fixed during posterior replay. The prior-mismatch replay replaces only the inference prior by a uniform spatial prior. True target positions retain the original regional mixture.

Table 18. Fixed-observation posterior diagnostics. Coverage is expressed in percent.

Prior and particles	$q_{.95}$ coverage	$q_{.99}$ coverage	Mean leakage bias
nominal_256	91.32 [84.49, 96.88]	95.86 [90.86, 99.34]	-0.01 [-0.07, 0.04]
nominal_1024	94.09 [88.33, 98.44]	98.78 [96.95, 100.00]	-0.02 [-0.08, 0.03]
uniform_256	74.69 [62.26, 85.42]	87.49 [77.78, 95.31]	-0.09 [-0.17, -0.03]
uniform_1024	78.12 [66.73, 88.09]	93.02 [85.89, 98.11]	-0.09 [-0.16, -0.03]

Table 19. Paired fixed-observation coverage differences in percentage points.

Contrast	$q_{.95}$ difference	$q_{.99}$ difference
nominal_1024 minus nominal_256	2.78 [-2.01, 7.86]	2.92 [0.27, 7.00]
uniform_1024 minus uniform_256	3.44 [-1.65, 9.47]	5.52 [1.56, 9.81]
uniform_256 minus nominal_256	-16.63 [-27.09, -7.48]	-8.36 [-14.89, -3.28]
uniform_1024 minus nominal_1024	-15.97 [-25.49, -7.88]	-5.76 [-12.22, -1.17]

A validation-only search over 32 missions selected an additive margin of 0.50 from the predeclared candidate set. The corrected margin is clipped at one and must not be interpreted as a posterior quantile. Fixed-margin options retain the original margin. Coverage improvement is evaluated together with mission cost. The candidate set is $\{0, 0.05, 0.10, 0.20, 0.30, 0.50\}$. Selection uses the original 256-particle seed and the smallest candidate whose pooled harvest coverage reaches 95% for both matched and uniform inference priors on validation trajectories. The rule would retain 0.50 and disclose failure if no candidate attained the target. At deployment a posterior-margin option applies $m = \min\{1, q_\alpha + \delta\}$, whereas a fixed-margin option retains its specified margin. The frozen model ranks all options using the changed observable features. No retraining follows calibration.

Table 20. Conservative full-policy transfer. Utility differences use the same scenarios as the uncorrected full policy.

Condition	Completion (%)	Utility difference	Padded candidate coverage (%)
nominal	100.00 [100.00, 100.00]	1.01 [-1.91, 5.44]	100.00 [100.00, 100.00]
uniform_inference	100.00 [100.00, 100.00]	2.00 [-1.75, 7.96]	98.44 [92.19, 100.00]
pd_mismatch	100.00 [100.00, 100.00]	3.51 [-0.63, 8.79]	100.00 [100.00, 100.00]
occlusion_mismatch	100.00 [100.00, 100.00]	0.34 [-4.43, 5.08]	100.00 [100.00, 100.00]

The frozen selector predominantly switches to fixed-margin options after correction. Therefore the improved coverage of padded candidate quantiles does not describe the margin applied by most selected options. The following table reports actual applied-margin coverage and energy cost separately.

Table 21. Applied-margin distinction and conservative-transfer cost. Both coverage columns are percentages.

Condition	Energy difference (kJ)	Original applied coverage	Corrected applied coverage
nominal	1.87 [0.14, 3.59]	90.44 [79.50, 98.44]	95.31 [87.50, 100.00]
uniform_inference	0.70 [-2.24, 3.80]	84.38 [73.44, 93.75]	90.62 [81.25, 98.44]
pd_mismatch	3.72 [1.77, 5.55]	81.21 [69.16, 92.15]	92.19 [84.38, 98.44]
occlusion_mismatch	2.08 [0.11, 4.10]	88.99 [79.31, 96.69]	92.19 [82.81, 98.44]

6.2 No-sensing comparison and building visibility stress

The strongest frozen no-sensing learner is compared with the full learner, gate, analytic selector and no-sensing budget portfolio. Six existing diagnostic conditions retain the original truth and exogenous streams. The additional visibility stress multiplies true detection probability by 0.2 when the public building raster blocks a ray to the ground target. Matched inference applies the same candidate-specific attenuation. Mismatched inference retains the distance-only likelihood. This stress changes the physical observation model and is not a prior-only mismatch. Original target cells, including cells within building footprints, are retained. The attenuation is a declared stress assumption without empirical calibration. For horizontal range d , the ray uses $N = \max\{\lfloor 2d/\Delta \rfloor, 2\}$ samples uniformly placed from 0.05 to 0.95 of the segment, where $\Delta = 40$ m is the grid width. A ray is blocked when any sampled building height exceeds the interpolated height between the UAV and the 1.5 m target. With visibility indicator v , the stressed detection model is $P_D^{\text{occ}} = P_D^{\text{range}}[v + 0.2(1 - v)]$. Matched inference computes v for each candidate particle location from the public raster. The hidden true location is used only by the observation generator. Matched and mismatched variants share the same detection uniforms and latent measurement noise at a common time and pose.

Table 22. Frozen-policy diagnostic transfer with mean and 95% interval.

Condition	Policy	Utility	Completion (%)	Secure payload (%)
nominal	full	105.57 [100.79, 108.80]	98.44 [93.75, 100.00]	98.58 [95.98, 99.85]
nominal	gate	105.57 [100.79, 108.80]	98.44 [93.75, 100.00]	98.58 [95.98, 99.85]
nominal	analytic	103.70 [99.42, 106.97]	98.44 [92.19, 100.00]	98.44 [95.97, 99.87]
nominal	budget_no_sensing	91.36 [82.56, 98.64]	100.00 [100.00, 100.00]	92.63 [87.53, 96.82]
nominal	no_sensing	98.23 [89.94, 104.75]	100.00 [100.00, 100.00]	94.01 [88.53, 98.26]
uniform_inference	full	102.48 [95.59, 107.38]	100.00 [100.00, 100.00]	96.76 [92.97, 99.25]
uniform_inference	gate	102.48 [95.59, 107.38]	100.00 [100.00, 100.00]	96.76 [92.97, 99.25]
uniform_inference	analytic	93.84 [84.47, 102.03]	100.00 [100.00, 100.00]	91.67 [86.05, 96.59]
uniform_inference	budget_no_sensing	90.28 [81.11, 97.72]	100.00 [100.00, 100.00]	91.99 [86.45, 96.49]
uniform_inference	no_sensing	99.44 [91.51, 106.08]	100.00 [100.00, 100.00]	93.43 [88.71, 97.35]
pd_matched	full	102.19 [97.34, 106.13]	100.00 [100.00, 100.00]	97.19 [94.08, 99.48]
pd_matched	gate	102.19 [97.34, 106.13]	100.00 [100.00, 100.00]	97.19 [94.08, 99.48]
pd_matched	analytic	102.37 [97.98, 105.89]	100.00 [100.00, 100.00]	97.67 [95.11, 99.51]
pd_matched	budget_no_sensing	91.36 [82.56, 98.64]	100.00 [100.00, 100.00]	92.63 [87.53, 96.82]
pd_matched	no_sensing	98.23 [89.94, 104.75]	100.00 [100.00, 100.00]	94.01 [88.53, 98.26]
pd_mismatch	full	101.82 [96.14, 106.43]	100.00 [100.00, 100.00]	96.49 [93.30, 98.94]
pd_mismatch	gate	101.82 [96.14, 106.43]	100.00 [100.00, 100.00]	96.49 [93.30, 98.94]

Table 22 continued

Condition	Policy	Utility	Completion (%)	Secure payload (%)
pd_mismatch	analytic	101.77 [96.74, 106.10]	100.00 [100.00, 100.00]	97.50 [94.83, 99.56]
pd_mismatch	budget_no_sensing	91.36 [82.56, 98.64]	100.00 [100.00, 100.00]	92.63 [87.53, 96.82]
pd_mismatch	no_sensing	98.23 [89.94, 104.75]	100.00 [100.00, 100.00]	94.01 [88.53, 98.26]
key_block1	full	5.44 [1.63, 9.39]	0.00 [0.00, 0.00]	23.25 [19.45, 27.16]
key_block1	gate	5.44 [1.63, 9.39]	0.00 [0.00, 0.00]	23.25 [19.45, 27.16]
key_block1	analytic	2.04 [-1.56, 6.27]	0.00 [0.00, 0.00]	18.79 [15.12, 23.20]
key_block1	budget_no_sensing	-4.37 [-6.37, -2.43]	0.00 [0.00, 0.00]	13.68 [12.61, 14.68]
key_block1	no_sensing	3.74 [1.80, 5.54]	0.00 [0.00, 0.00]	14.53 [13.47, 15.62]
key_block16	full	107.15 [102.28, 110.78]	100.00 [100.00, 100.00]	98.32 [95.74, 100.00]
key_block16	gate	99.16 [91.17, 105.92]	100.00 [100.00, 100.00]	95.16 [90.82, 98.63]
key_block16	analytic	106.80 [101.70, 110.22]	100.00 [100.00, 100.00]	98.84 [96.59, 100.00]
key_block16	budget_no_sensing	94.48 [85.70, 101.96]	100.00 [100.00, 100.00]	92.63 [87.53, 96.82]
key_block16	no_sensing	109.54 [106.55, 112.33]	100.00 [100.00, 100.00]	99.85 [99.44, 100.00]
occlusion_matched	full	102.24 [96.51, 106.03]	100.00 [100.00, 100.00]	97.67 [94.20, 99.79]
occlusion_matched	gate	102.24 [96.51, 106.03]	100.00 [100.00, 100.00]	97.67 [94.20, 99.79]
occlusion_matched	analytic	102.15 [96.61, 106.07]	100.00 [100.00, 100.00]	97.75 [94.33, 99.82]
occlusion_matched	budget_no_sensing	91.36 [82.56, 98.64]	100.00 [100.00, 100.00]	92.63 [87.53, 96.82]
occlusion_matched	no_sensing	98.23 [89.94, 104.75]	100.00 [100.00, 100.00]	94.01 [88.53, 98.26]
occlusion_mismatch	full	103.79 [98.39, 107.68]	98.44 [93.75, 100.00]	97.95 [95.19, 99.63]
occlusion_mismatch	gate	103.79 [98.39, 107.68]	98.44 [93.75, 100.00]	97.95 [95.19, 99.63]
occlusion_mismatch	analytic	101.35 [95.30, 105.75]	100.00 [100.00, 100.00]	96.97 [93.34, 99.48]
occlusion_mismatch	budget_no_sensing	91.36 [82.56, 98.64]	100.00 [100.00, 100.00]	92.63 [87.53, 96.82]
occlusion_mismatch	no_sensing	98.23 [89.94, 104.75]	100.00 [100.00, 100.00]	94.01 [88.53, 98.26]

Table 23. Paired utility differences for the strongest no-sensing learner. Positive differences favor no sensing.

Condition	No sensing minus full	No sensing minus gate
nominal	-7.34 [-16.00, 0.73]	-7.34 [-16.00, 0.73]
uniform_inference	-3.04 [-10.38, 3.58]	-3.04 [-10.38, 3.58]
pd_matched	-3.96 [-11.88, 3.55]	-3.96 [-11.88, 3.55]
pd_mismatch	-3.58 [-12.77, 5.50]	-3.58 [-12.77, 5.50]
key_block1	-1.70 [-6.15, 2.90]	-1.70 [-6.15, 2.90]
key_block16	2.39 [-1.01, 6.44]	10.38 [2.90, 19.20]
occlusion_matched	-4.01 [-12.88, 4.17]	-4.01 [-12.88, 4.17]
occlusion_mismatch	-5.56 [-14.45, 2.39]	-5.56 [-14.45, 2.39]

A budget-guard event counts an intercepted unaffordable return action. The counter does not imply executed energy overshoot. Trajectory-conditioned coverage can change when a policy visits different cells, whereas the fixed-observation replay holds that source of variation constant. Increasing particle count alone does not establish posterior exactness or eliminate local resampling and rejuvenation error.

7 Detailed model scope and units

The following sections specify the simulator underlying the operational account in Section 2 and the evaluations in Sections 4 through 6. The sequence follows scene geometry, channel conditions, key accounting, sensing, authentication, energy, and learning inputs. Distances are in metres, time is in seconds, energy is in kilojoules, payload is in megabits, and key inventory is in kilobits unless another unit is stated. Decimal prefixes apply. One megabit is 10^6 bits and one kilobit is 10^3 bits. The physical model is the frozen V4 model. V5 adds an initial sensing gate, comparison policies, fresh confirmation scenes, and diagnostic interventions. V6 adds a learned skip branch, feature variability, compact models, a second confirmation cohort, and controlled authentication and inference diagnostics. Changes from the frozen physical model are confined to explicitly named interventions.

The implementation contains motion, channel maps, a particle belief, gateway-specific inventories, a service freshness interface, mission selection, and execution checks. Channel and key rates are parametric proxies. Radio and propulsion fingerprints do not have implemented feature extractors or classifiers. The common authentication interface represents freshness-dependent rejection probabilities. Protected telemetry, credential provisioning, trustworthy gateways, and secure local inventory storage are deployment assumptions. Platform diversity and cooperative swarms are future extensions. No flight measurements calibrate the functions below.

8 Scene and operational state

The environment contains 25×25 cells with a cell edge of 40 m. A cell with column c and row r has centre $(40(c + 0.5), 40(r + 0.5))$. The altitude index $a \in \{0, 1, 2\}$ represents $h_a \in \{60, 100, 140\}$ m. The initial altitude is 100 m. Each action occupies $\Delta t = 4$ s, and the horizon is $T_{\max} = 200$ actions.

The initial horizontal cell is (2, 21) and the recovery cell is (22, 2). Gateway A is at cell (20, 4) and height 25 m. Gateway B is at cell (4, 18) and height 12 m. Map generation places 26 rectangular building blocks with side lengths of one to three cells and heights uniformly sampled from 15 to 60 m. Overlapping blocks retain the maximum height. The start, recovery, and gateway cells are cleared. The simulation treats building geometry as an input to radio visibility. The executor does not implement continuous collision avoidance or a landing trajectory.

The observable operational state contains position, altitude, time index t , remaining energy E_t , remaining payload D_t , total key inventories $K_{g,t}$, the receiver posterior, and authentication freshness. Initially $D_0 = 400$ Mbit, both key pools are empty, and $E_0 \in \{70, 90, 110, 130\}$ kJ. The passive receiver count $K \in \{1, 2, 3\}$ and ground positions form a hidden state X . The evaluator additionally stores unexposed and exposed inventory portions. Neither the hidden receiver state nor the inventory split enters the policy interface.

There are eleven primitive actions. Four actions move by one cell in a cardinal direction. Two actions change altitude by one level. The remaining actions generate keys, transmit payload, sense, authenticate, or hover. Movement is clipped at grid boundaries. Horizontal and vertical speeds are nominally 10 m/s. A service action and a movement action cannot occur in the same interval.

9 Channel and visibility maps

For a UAV position p and gateway g , let $d_{3,g}$ be the three-dimensional separation. A ray is sampled from 5% to 95% of the segment at approximately half-cell spacing. The link is blocked if any sampled building exceeds the ray height. Let $L_g = 1$ denote an unblocked ray. This deterministic test supplies the visibility state for the channel map.

The path loss in dB and the signal-to-noise ratio are

$$P_{L,g} = 20 \log_{10}(\max(d_{3,g}, 1)) + 20 \log_{10}(2.4 \times 10^9) - 147.55 + \eta_g + S(c, r), \quad (2)$$

$$\eta_g = L_g \times 1 + (1 - L_g) \times 20, \quad (3)$$

$$N_{\text{dBm}} = -174 + 10 \log_{10}(10^6) + 7 = -107, \quad (4)$$

$$\gamma_{g,\text{dB}} = 20 - P_{L,g} - N_{\text{dBm}}, \quad \gamma_g = 10^{\gamma_{g,\text{dB}}/10}. \quad (5)$$

Here the transmit power is 20 dBm, carrier frequency is 2.4 GHz, receiver noise figure is 7 dB, and data bandwidth is 1 MHz. The shadow field S is generated by applying a Gaussian spatial filter with a three-cell standard deviation to white Gaussian samples. The resulting field is normalized to a sample standard deviation of 4 dB. The same field value is used at the same horizontal cell for every gateway and altitude. Thus 120 m specifies the smoothing scale. The code does not generate independent shadow fields for different links.

The payload-rate proxy is

$$R_g(p) = \log_2(1 + \gamma_g(p)) \quad \text{Mbit/s.} \quad (6)$$

This Shannon-rate proxy excludes modulation selection, retransmissions, and packet overhead. Configured probabilistic line-of-sight coefficients and Rician factors are inactive in this implementation. No stochastic Rician fading or elevation-dependent line-of-sight probability is evaluated. A comment referring to 12 MHz probing bandwidth does not define an active bandwidth parameter.

10 Key generation and exposure accounting

10.1 Gross yield and assumed reconciliation loss

The attitude-jitter parameter is $\sigma_a \in \{1.2, 2.0, 3.2\}$ degrees for the three flight levels. The assumed coherence time, sampling frequency, reciprocal correlation, and effective bits per sample are

$$T_c(a) = \frac{0.050}{1 + \sigma_a/4} \quad \text{s,} \quad (7)$$

$$f_s(a) = \min\{100, 1/T_c(a)\} \quad \text{Hz,} \quad (8)$$

$$\rho_g(p) = \frac{\gamma_g(p)}{1 + \gamma_g(p)} \exp\left[-\left(\frac{0.005}{T_c(a)}\right)^2\right], \quad (9)$$

$$b_g(p) = 2 \max\left\{0, \frac{\rho_g(p) - 0.93}{1 - 0.93}\right\}. \quad (10)$$

The factor of two is an assumed quantizer capacity. The thresholded linear function represents effective retained observations rather than an implemented bit quantizer. With eight assumed independent subbands, the available gross rate is

$$G_g(p) = 8f_s(a)b_g(p) \mathbf{1}\{d_{U,g} \leq 340\} \quad \text{bit/s,} \quad (11)$$

where $d_{U,g}$ is horizontal separation from the selected gateway. The 340 m limit is a hard probing-availability assumption. This limit is distinct from the smooth receiver-exposure model below. There is no finite distance beyond which receiver leakage is forced to zero.

A fixed factor $\eta_{\text{rec}} = 0.8$ represents reconciliation retention. No Cascade or BCH transcript is simulated despite the historical configuration name `cascade_overhead`. No raw bit-disagreement measurements, finite-block error probability, or implemented privacy hash is available. The coherence, jitter, and rate functions therefore remain illustrative assumptions.

10.2 Receiver-dependent exposure and chosen margin

Let e_k be the horizontal location of passive receiver k . Let p_{xy} and g_{xy} be the horizontal locations of the UAV and gateway. The assumed short-scale unexposed share is

$$w_g(p) = \min\{0.9, w_{0,g} + 0.08\sigma_a\}, \quad (12)$$

$$w_{0,g} = 0.15L_g + 0.60(1 - L_g). \quad (13)$$

The individual and aggregate exposure fractions are

$$d_{k,g}(p) = \min\{\|p_{xy} - e_k\|, \|g_{xy} - e_k\|\}, \quad (14)$$

$$u_{k,g}(p) = (1 - w_g(p)) \exp[-(d_{k,g}(p)/200)^2], \quad (15)$$

$$\ell_g(p, X) = 1 - \prod_{k=1}^K (1 - u_{k,g}(p)). \quad (16)$$

These expressions use horizontal receiver distances. Altitude affects visibility and the jitter parameter, rather than entering the distance kernel directly. The product combines proxy exposed shares and does not describe arbitrary colluding receivers or information-theoretic side information.

The belief consists of particles X_j with weights ω_j . A posterior-quantile margin is the smallest weighted empirical quantile

$$m_q(p, g) = \inf \left\{ z \left| \sum_j \omega_j \mathbf{1}\{\ell_g(p, X_j) \leq z\} \geq q \right. \right\}. \quad (17)$$

The supported quantiles are $q = 0.95$ and 0.99 . Fixed-margin profiles use $m = 0.45, 0.65$, or 0.80 . The expected residual fraction used for planning is

$$\bar{e}(p, g, m) = \sum_j \omega_j \max \left\{ 0, \frac{\ell_g(p, X_j) - m}{\max(1 - m, 10^{-12})} \right\}. \quad (18)$$

The nonlinear residual is averaged over the posterior. The planner does not substitute the mean or a quantile into this expression and call the result expected exposure.

10.3 Gateway pools and payload accounting

During one key-generation interval, the retained key amount before the pool cap is

$$Y_g = \frac{G_g(p)\Delta t}{1000} (1 - m) \eta_{\text{rec}}, \quad (19)$$

$$e_g^* = \max \left\{ 0, \frac{\ell_g(p, X^*) - m}{\max(1 - m, 10^{-12})} \right\}, \quad (20)$$

$$Y_g^{\text{exp}} = Y_g e_g^*, \quad Y_g^{\text{unexp}} = Y_g (1 - e_g^*). \quad (21)$$

The asterisk denotes evaluator truth. The two portions are multiplied by

$$s_g = \min \left\{ 1, \frac{\max(0, 30 - K_g)}{\max(Y_g, 10^{-12})} \right\} \quad (22)$$

before being added to the selected pool. The pool cap is 30 kbit per gateway. Policies observe $K_g = K_g^{\text{unexp}} + K_g^{\text{exp}}$ and cannot inspect either component separately.

A fresh 256-bit key covers up to $B = 4$ Mbit in the nominal workload. This is a frequent-rekeying assumption and is not one-time-pad encryption. At transmission, the available complete key blocks and payload amount are

$$n_{K,g} = \left\lfloor \frac{1000 K_g}{256} \right\rfloor, \quad (23)$$

$$v_g = \min\{R_g(p)\Delta t, D_t, B n_{K,g}\}, \quad (24)$$

$$k_{\text{used}} = 0.256 \left\lceil \frac{v_g}{B} \right\rceil \quad \text{kbit}. \quad (25)$$

The implementation uses small floating-point tolerances at integer boundaries. For nonzero v_g , the pre-transmission unexposed ratio is $r_g = K_g^{\text{unexp}}/K_g$. Both inventory portions are depleted proportionally. The payload counters increase by $v_g r_g$ and $v_g (1 - r_g)$ for modeled secure and modeled exposed payload, respectively. Remaining payload decreases by v_g .

An ideal unfragmented 400 Mbit transfer at $B = 4$ Mbit needs 25.6 kbit. Partial transmissions can consume more blocks through the ceiling operation. The continuous inventory weighting is an evaluator accounting convention. It does not establish packet-level compromise events or a composable secrecy bound. The geometric margin is not a substitute for a conditional entropy bound required by privacy amplification.

11 Sensing and posterior approximation

11.1 Hidden population and prior

The true count is uniform on one, two, and three receivers. Receiver cells are drawn independently from a spatial prior with repeated draws rejected until all cells differ. Both the true scene and nominal inference use

$$\pi(c, r) = 0.98 \frac{\mathbf{1}\{(c, r) \in \mathcal{D}\}}{|\mathcal{D}|} + \frac{0.02}{625}. \quad (26)$$

The set $\mathcal{D}$ is the union of inclusive cell rectangles (14, 0) to (24, 10) and (0, 12) to (10, 22). Each receiver has height 1.5 m and remains static. The floor permits out-of-district hypotheses but does not guarantee that any finite set of measurements can identify them.

11.2 Detection, localization, and clutter

For horizontal separation d at altitude h , the slant range is $s = \sqrt{d^2 + (h - 1.5)^2}$. The nominal support and detection probability are

$$R_{\text{sense}}(h) = 520(h/100)^{0.25}, \quad (27)$$

$$P_D(s, h) = \frac{0.95}{2} [1 + \cos(\pi \min\{s/R_{\text{sense}}(h), 1\})]. \quad (28)$$

Each receiver can generate at most one return. Conditional on detection, the measured horizontal position is Gaussian in local radial and tangential coordinates. The standard deviations are

$$\sigma_r(s) = 12 + 0.10s \quad \text{m}, \quad (29)$$

$$\sigma_\theta(s) = (3 + 0.012s)\pi/180 \quad \text{rad}, \quad (30)$$

$$\sigma_\perp(d, s) = \max\{10, d\sigma_\theta(s)\} \quad \text{m}. \quad (31)$$

The covariance in horizontal coordinates is

$$\Sigma = \sigma_r^2 e_r e_r^\top + \sigma_\perp^2 e_\perp e_\perp^\top, \quad (32)$$

where e_r and $e_\perp$ are radial and tangential unit vectors. An additional clutter return occurs with probability 0.02 and is uniform in the square city. Measurements are shuffled and do not reveal target identity. Receiver positions determine simulated observations only. The inference code evaluates candidate-dependent detection probabilities for candidate positions.

11.3 Unlabelled measurement likelihood

Let $Z = \{z_1, \dots, z_n\}$ be a dwell's observed set. For a hypothesized target set X , an admissible association A maps a subset of targets injectively to observations. Let $C_A \in \{0, 1\}$ count observations left for clutter. Associations leaving more than one clutter observation are inadmissible. With Gaussian density $f_k(z)$ and uniform clutter density $c(z) = 10^{-6}$ within the city, the likelihood is

$$\mathcal{L}(Z | X) = \sum_A (0.02)^{C_A} (0.98)^{1-C_A} \prod_{(k,j) \in A} P_{D,k} f_k(z_j) \prod_{k \notin A} (1 - P_{D,k}) \prod_{j \in \text{clutter}(A)} c(z_j). \quad (33)$$

The code evaluates this finite sum by dynamic programming over a used-target mask and a binary clutter flag. Likelihoods are floored at 10^{-300} to avoid numerical underflow. An empty return set therefore remains informative through missed-detection probabilities.

The filter contains 256 particles and assigns initial total weight 1/3 to each count stratum. The number of particles per stratum differs by at most one. Initial locations follow the distinct-cell prior. After a dwell,

$$\omega_j^+ = \frac{\omega_j \mathcal{L}(Z | X_j)}{\sum_i \omega_i \mathcal{L}(Z | X_i)}. \quad (34)$$

Resampling is performed separately within a count stratum when the conditional effective sample size falls below half that stratum’s size. The total posterior weight of each count is preserved. Each resampled stratum receives one Metropolis rejuvenation sweep. The proposal moves one randomly selected receiver by independently sampled integer offsets from -2 to 2 cells in both directions. Coordinates wrap modulo 25. Duplicate active receiver cells are rejected. The acceptance ratio evaluates the prior and every accumulated dwell likelihood. These numerical proposals are a posterior exploration mechanism, not physical receiver motion.

The plotted occupancy distribution is

$$b(c, r) = \frac{\sum_j \omega_j \sum_{k=1}^{K_j} \mathbf{1}\{e_{jk} = (c, r)\}}{\sum_j \omega_j K_j}. \quad (35)$$

Its entropy is $H_b = -\sum_{c,r} b(c, r) \log b(c, r)$ in nats. This is the entropy of the normalized occupancy distribution rather than entropy of the complete random-set posterior. Feature 12 separately divides this entropy by $\log 625$. The estimated count is $\hat{K} = \sum_j \omega_j K_j$.

12 Authentication, attacker capabilities, and event order

Forged frames arrive according to a stationary Poisson process with mean 0.08 per four-second interval. The corresponding rate is 0.02 frames/s. The attacker can submit forged frames for classification by the simulated interface. A frame marked accepted does not modify flight commands, sensor values, inventories, or gateway state. The simulator does not implement credential verification, replay protection, hardware imitation, a challenge message format, or cryptographic command processing.

Freshness F_t is an integer countdown initialized to six. An authentication action uses the highest gateway SNR at the current position, rather than necessarily the selected service gateway. A challenge succeeds with probability 0.98 when that SNR is at least 5 dB and with probability 0.60 otherwise. Success resets freshness to six and age to zero. Failure leaves the countdown unchanged.

After the selected action’s effect, forged frames for that same interval are evaluated. For $N_t \sim \text{Poisson}(0.08)$,

$$p_{\text{acc}}(F_t) = \begin{cases} 0.005, & F_t > 0, \\ 0.70, & F_t = 0, \end{cases} \quad (36)$$

$$A_t \mid N_t, F_t \sim \text{Binomial}(N_t, p_{\text{acc}}(F_t)). \quad (37)$$

The rejected count $J_t = N_t - A_t$ is observable. The anomaly memory is $M_t = 0.8M_{t-1} + J_t$. The accepted count is evaluator-only. The interval ends by decreasing freshness by one to a minimum of zero and increasing authentication age by one. Thus a successful challenge already protects the interval in which the challenge occurs.

All mission options use the same threshold and request authentication when freshness is at most one. This rule does not depend on anomaly memory and is not learned. Returning with no payload or inadequate energy takes priority over this request. A separate service guard replaces attempted key generation or transmission by authentication when freshness has already expired. The reported proxy acceptance rate is total accepted frames divided by total attempted frames. Different mission durations and stale recovery intervals can change this ratio without changing a classifier.

The spoof-arrival, acceptance-uniform, and challenge-uniform streams are indexed by mission time. Sensing randomness is separately indexed by map seed, mission index, and time. Earlier calls cannot shift a later attack sequence. Policies compared in the same scene therefore encounter a common exogenous process at matching times.

Authenticated bootstrap credentials, trustworthy channel maps and sensor reports, protected local inventories, and uncompromised gateways are assumed. Active pilot contamination, manipulated sensing returns, malicious telemetry, compromised gateways, and arbitrary collusion are excluded. Detecting a physical object does not establish that the object is malicious. Undetectable passive receivers remain outside any coverage guarantee.

13 Energy accounting and recovery

For horizontal speed v , the rotary-wing propulsion approximation is

$$P(v) = P_0(1 + 3v^2/U_{\text{tip}}^2) + P_i \left(\sqrt{1 + v^4/(4v_0^4)} - v^2/(2v_0^2) \right)^{1/2} + \frac{1}{2}d_0\rho_{\text{air}}s_RA_Rv^3. \quad (38)$$

The parameters are $P_0 = 79.86$ W, $P_i = 88.63$ W, $U_{\text{tip}} = 120$ m/s, $v_0 = 4.03$ m/s, $d_0 = 0.6$, $\rho_{\text{air}} = 1.225$ kg/m³, rotor solidity $s_R = 0.05$, and rotor area $A_R = 0.503$ m². This horizontal-flight expression follows the rotary-wing model cited in the main article. The climb and descent terms below are separate approximations.

Hover power is $P(0) = 168.49$ W. A cardinal move costs $e_M = P(10)\Delta t/1000$ kJ. Key generation, transmission, and authentication each cost $e_C = (P(0) + 5)\Delta t/1000 = 0.69396$ kJ. Sensing costs $e_S = (P(0) + 18)\Delta t/1000 = 0.74596$ kJ. An idle hover costs 0.67396 kJ. A 40 m climb costs

$$e_{\uparrow} = \frac{P(0)\Delta t + 3 \times 9.81 \times 40/0.7}{1000} = 2.35567 \quad \text{kJ}. \quad (39)$$

A descent costs $e_{\downarrow} = 0.6P(0)\Delta t/1000 = 0.404376$ kJ. Climb and descent actions do not add communication or sensing power. The 18 W sensing increment is a lumped dwell and processing assumption. A dedicated measurement of general computation, idle sensing electronics, protocol retransmissions, and flight-controller auxiliary power is absent. Offline training energy and desktop inference energy do not enter mission energy.

For a requested action, the environment first applies the freshness service check. The recovery guard then predicts the action cost $e(a)$ and next horizontal cell p' . The action is permitted only if

$$E_t \geq e(a) + e_M\|p' - p_{\text{rec}}\|_1 + E_{\text{res}}, \quad (40)$$

where the Manhattan distance is in cells and nominal $E_{\text{res}} = 1$ kJ. Otherwise the guard enters a persistent return mode and requests a cardinal step toward recovery. The guard does not include landing energy, obstacle avoidance, wind uncertainty, or certified descent. If the next return step is unaffordable, the mission terminates before that step is executed. The historical budget_violations counter also counts this event and therefore does not directly measure actual energy overshoot.

Strict completion requires all payload delivered, arrival at the recovery cell, and compliance with the energy budget. It does not additionally require zero modeled exposure or zero accepted frames. A recovered incomplete mission receives no completion bonus. Modeled secure percentage is 100 times secure Mbit divided by 400. Completion-adjusted modeled secure payload multiplies secure Mbit by the strict-completion indicator. Aggregate task energy efficiency divides the total completion-adjusted secure Mbit by total energy across all evaluated missions, including failed missions.

14 Mission options and the common analytic forecast

14.1 The option catalogue

An option specifies gateway g , target altitude index a_o , privacy mode, and additional sensing allowance s_o . There are $2 \times 3 \times 6 = 36$ options. The six profiles are

Privacy rule	Margin or quantile	Additional dwells
Fixed	0.45	0
Fixed	0.65	0
Fixed	0.80	0
Posterior quantile	0.95	0
Posterior quantile	0.95	2
Posterior quantile	0.99	4

The authentication threshold is one for every option. Initial sensing, when selected, consists of two additional paid dwells before option selection. The six profiles couple privacy and sensing in the main option catalogue. The separate crossed experiment removes that coupling by independently crossing three privacy rules with zero, two, and four additional dwells at a fixed gateway and altitude.

14.2 Candidate waypoints and resource forecasts

For each gateway and target altitude, candidate harvest sites are the 12 cells with highest positive gross key rate. The current and recovery cells are added when their gross rate is positive. Ties use stable array order. Candidate transmission sites are the four cells with highest positive payload rate. For each harvest candidate, that harvest cell and the recovery cell are also considered as transmission sites.

For a harvest site H and transmission site Q , let y_H be retained key yield per dwell under the chosen margin. The approximate transmission, key, harvest, and travel requirements are

$$\hat{n}_T = \lceil D_t / (R_g(Q) \Delta t) \rceil, \quad (41)$$

$$\hat{n}_B = \lceil D_t / B \rceil + \max(0, \hat{n}_T - 1), \quad (42)$$

$$\hat{k} = \max(0, 0.256 \hat{n}_B - K_g), \quad (43)$$

$$\hat{n}_H = \lceil \hat{k} / y_H \rceil, \quad (44)$$

$$\hat{n}_M = \|p - H\|_1 + \|H - Q\|_1 + \|Q - p_{\text{rec}}\|_1, \quad (45)$$

$$\hat{n}_V = |a_o - a_t|, \quad (46)$$

$$\hat{n}_W = \hat{n}_H + \hat{n}_T + \hat{n}_M + \hat{n}_V + s_o, \quad (47)$$

$$\hat{n}_A = \lceil \hat{n}_W / 4 \rceil, \quad \hat{n} = \hat{n}_W + \hat{n}_A. \quad (48)$$

The extra key-block term conservatively allows for partial transmissions. The authentication count is an approximate forecast from the common threshold rule. It is not an exact replay of challenge successes or the countdown. The forecast charges the selected additional sensing allowance and includes recovery travel. Future sensing observations are not simulated by the analytic predictor.

The predicted energy and completion indicator are

$$\hat{E} = \hat{n}_H e_C + \hat{n}_T e_C + \hat{n}_M e_M + \hat{E}_V + \hat{n}_A e_C + s_o e_S, \quad (49)$$

$$\hat{I} = \mathbf{1}\{\hat{E} \leq E_t\} \mathbf{1}\{\hat{n} \leq 200 - t\}, \quad (50)$$

where $\hat{E}_V$ sums the corresponding climb or descent costs. Within each option, the chosen candidate pair minimizes

$$C(H, Q) = \hat{E} + 0.05 \hat{n} + 60 \bar{e}(H, g, m). \quad (51)$$

The coefficients define a fixed planning preference. They convert delay and residual exposure to a cost scale and are not physical energy constants. A site with negligible retained yield is skipped. If no pair remains, the implementation returns a large-cost placeholder of 10^6 kJ and 10^6 steps with unit exposure. Every compared option-ranking method calls this same forecast.

Define the feasible delivery fraction and expected fresh-state false acceptances as

$$f = \min \left\{ 1, \frac{1}{\max\{\hat{E}/E_t, \hat{n}/\max(1, 200 - t), 10^{-9}\}} \right\}, \quad (52)$$

$$\hat{A} = 0.08 \hat{n} (1 - 0.995). \quad (53)$$

The frozen analytic option value is

$$V_{\text{an}} = 100f(1 - \bar{e}) + 20\hat{I} - 70f\bar{e} - 5\hat{A} - 0.10\hat{E}. \quad (54)$$

This score assumes fresh-state rejection in the forecast. It does not model the exact stale recovery period. Residual learning can correct systematic forecast error in training, but no general correction guarantee follows.

14.3 Common executor pseudocode

The following ordering is part of the implementation. Replanning means recomputing service waypoints for the selected option. Ordinary execution does not change the gateway, target altitude, privacy rule, or initial additional sensing allowance.

1. Set the selected gateway and privacy rule in the service interface.
2. Compute horizontal return energy plus an internal 1 kJ margin. If no payload remains or available energy is at most that amount, request a horizontal return step.
3. Otherwise, if freshness is at most one, request authentication.
4. Otherwise, if the additional sensing allowance is not exhausted, at least eight intervals have elapsed since the last policy sensing request, and energy exceeds return need plus sensing cost plus 2 kJ, request sensing and invalidate the service plan.
5. Recompute the analytic waypoint plan if no plan exists, the posterior revision changed, or remaining payload changed. A key-pool change alone does not trigger this replan.
6. Set the desired pool to the forecast remaining key demand, capped at $0.95 \times 30 = 28.5$ kbit. If the selected gateway pool is below that amount, use the harvest waypoint. Otherwise use the transmission waypoint.
7. Move vertically toward the selected altitude first, then horizontally in the column direction, then in the row direction. At the chosen waypoint, request key generation or transmission.
8. Apply the environment freshness and return-energy guards to the requested action. Execute the admitted action and record both the request and the executed action.

The planner’s selected sensing allowance is a maximum. Early completion, recovery, or inadequate residual energy can prevent all requested additional dwells from being used. In the rolling analytic reference, the remaining-mission option forecast is repeated every eight intervals. Each candidate forecast subtracts already used additional dwells from the candidate allowance, with a minimum of zero. A new option replaces the current option only when the new analytic score exceeds the current score by more than two utility units. The executor retains the mission-wide sensing count and last sensing time after a switch. This rolling comparison is separate from the learned mission-option executor.

15 Learning targets and observable feature definitions

15.1 Fixed task objective and option regression

For a completed simulator execution, let S and L be the secure and exposed percentages of the offered payload, I the strict-completion indicator, A the accepted-frame count, and E the consumed energy. The fixed training and evaluation utility is

$$U = S + 20I - 0.7L - 5A - 0.10E. \quad (55)$$

The direct predictor regresses U on the 37-dimensional option feature vector $\phi(x, o)$. The residual predictor regresses $U - V_{\text{an}}(x, o)$ and uses

$$\hat{V}(x, o) = V_{\text{an}}(x, o) + \hat{f}_{\text{RF}}(\phi(x, o)) \quad (56)$$

for ranking. The original full sensing and separately trained no-sensing selectors each have 256 trees and a minimum leaf size of three. The V6 compact selector has 32 trees and a minimum leaf size of eight. Training labels are complete observed simulator outcomes. No Bellman iteration, temporal-difference target, or policy-gradient update occurs. Per-step reward-shaping quantities retained in the simulator are not the regression target.

15.2 The 37 features

Every feature is dimensionless. Let E_0 be the initial budget, a the current altitude index, g_o the candidate gateway index, and a_o the candidate altitude index. Let M_0, M_1 be normalized occupancy mass in the two prior districts. Let $\mu_x, \mu_y, \sigma_x, \sigma_y$ be moments of the occupancy distribution. Quantities with a hat come from the common forecast. Numerical denominator floors match the implementation.

Table 24. Exact one-based feature ordering.

Index	Meaning	Value
1	Constant	1
2	Remaining energy fraction	E_t/E_0
3	Initial budget scale	$E_0/130$
4	Remaining payload fraction	$D_t/400$
5	Elapsed fraction	$t/200$
6	Column coordinate	$c/25$
7	Row coordinate	$r/25$
8	Current altitude index	$a/2$
9	Gateway A inventory	$K_0/30$
10	Gateway B inventory	$K_1/30$
11	Freshness fraction	$F_t/6$
12	Occupancy entropy	$H_b/\log 625$
13	First district mass	M_0
14	Second district mass	M_1
15	Estimated receiver count	$\hat{K}/3$
16	Occupancy mean x coordinate	$\mu_x/1000$
17	Occupancy mean y coordinate	$\mu_y/1000$
18	Occupancy x spread	$\sigma_x/1000$
19	Occupancy y spread	$\sigma_y/1000$
20	Selected gateway	$g_o \in \{0, 1\}$
21	Selected altitude index	$a_o/2$
22	Fixed-margin indicator	$\mathbf{1}\{\text{fixed privacy mode}\}$
23	Fixed margin	The fixed m or zero for a quantile profile
24	Quantile parameter	q , with the unused default 0.95 retained for fixed profiles
25	Additional sensing scale	$s_o/4$
26	Predicted energy ratio	$\min\{\hat{E}/\max(E_t, 10^{-9}), 20\}$
27	Predicted time ratio	$\min\{\hat{n}/\max(1, 200 - t), 20\}$
28	Predicted energy slack	$\text{clip}((E_t - \hat{E})/E_0, -20, 1)$
29	Mean exposure at harvest site	$\sum_j \omega_j \ell_g(H, X_j)$
30	Quantile exposure at harvest site	$m_q(H, g)$
31	Applied harvest margin	Fixed m or posterior $m_q(H, g)$
32	Expected residual exposure	$\bar{e}(H, g, m)$
33	Predicted full completion	$\hat{I}$
34	Retained key yield scale	$y_H/2$, with y_H in kbit per interval
35	Remaining key demand scale	$\hat{k}/25.6$, with $\hat{k}$ in kbit
36	Travel count scale	$\hat{n}_M/100$
37	Authentication count scale	$\min\{\hat{n}_A/50, 20\}$

The normalized coordinate features use cell indices rather than cell-centre distances. The 25.6 kbit normalization remains fixed in block-size diagnostics. The posterior mean and quantile in features 29 and 30 are retained even for a fixed-margin option. These values are observable forecasts, not hidden receiver truth. Authentication age, rejected-frame memory, true compromise labels, map identifiers, and mission identifiers are excluded.

15.3 The original 451-feature initial sensing gate

Before any sensing action, construct all 36 option feature vectors. The gate concatenates features 1 through 19 from the first option with features 26 through 37 from each option in catalogue order. Thus the input has $19 + 36 \times 12 = 451$ entries. Catalogue order places gateway first, altitude second, and the six profiles last. The gate input contains current observable summaries and analytic forecasts. No future observation or simulator-only outcome label enters the gate input.

For the original V5 comparator, the two branches are a frozen validation-selected budget portfolio with no initial sensing and two paid sensing dwells followed by residual option selection. Gate targets are paired differences in realized utility between the branches. To reduce in-sample option-selection optimism, training maps are partitioned into four folds. A residual selector is fitted on three folds and selects the sensing branch’s option outcomes for the held-out maps. All missions, budgets, and candidate options from a given map remain together. These held-out branch outcomes supply gate training labels. The skip branch uses the frozen budget portfolio’s actual zero-prefix outcomes.

A random forest predicts the branch advantage $\hat{\Delta}$. Candidate gate leaf sizes are 3, 8, and 16. Candidate thresholds are 0, 2, 5, and 10 utility units. Validation also considers always skipping. The gate requests the two dwells only if $\hat{\Delta}$ exceeds the selected threshold, unless validation selects always skipping. The paid branch then uses the frozen residual selector fit zero. The skip branch directly chooses the budget portfolio option. The gate is evaluated with one prespecified fit and should not be described as having four independent gate fits.

The initial V4 test had already been inspected before the V5 comparisons were developed. V5 subsequently introduced cohort A. V6 development could inspect cohort A and therefore uses the separate cohort B for confirmation. Each confirmation cohort has eight maps, 20 scenarios per map, and four budgets. The 640 contexts represent 160 scenario clusters. Earlier four-fit results do not add independent fits to either confirmation cohort.

16 Diagnostic interventions and interpretation

An accounting-only comparison evaluates three residual-exposure transformations on the same confirmation trajectories. For exponent $\alpha \in \{0.5, 1, 2\}$, a shadow inventory assigns the retained key material exposed fraction $(e_g^*)^\alpha$ at harvest. Each shadow pool subsequently follows the same proportional depletion rule as the nominal pool. Only the evaluator’s exposed and unexposed labels differ. Observations, total available keys, action sequence, delivered payload, energy, and completion are unchanged. Utility is then recomputed using the corresponding shadow secure and exposed payload amounts. This comparison measures dependence on the residual-exposure accounting rule and does not retune a policy or validate secrecy.

The physical-mapping diagnostics vary the payload covered by each 256-bit key across 1, 4, and 16 Mbit. The ideal unfragmented key demands are consequently 102.4, 25.6, and 6.4 kbit for the same payload. A second intervention changes the exposure distance scale from 200 m to 150 or 300 m. A third substitutes

$$\ell_g^{\text{alt}}(p, X) = 1 - \exp \left[- \sum_{k=1}^K u_{k,g}(p) \right] \quad (57)$$

for the nominal product mapping. Policies are frozen, but the common forecast reads the changed public physical configuration. These are known-parameter mapping sensitivities, not hidden distribution shifts and not empirical calibration.

Inference-mismatch diagnostics preserve the same true scene. A uniform inference prior replaces the regional inference prior without changing the actual receiver sampling distribution. Another condition lowers the true maximum detection probability to 0.60 while inference still assumes 0.95. A matched 0.60 filter supplies the comparison under that same changed sensor. A localization-noise condition multiplies true radial and tangential standard deviations by 1.8 while inference retains nominal noise. A matched 1.8 filter supplies the corresponding comparison. Matched and mismatched inference should be compared under common generated observations. Changing the actual receiver distribution instead would answer a different question.

Recovery-margin diagnostics set the environment guard reserve to 0, 1, or 3 kJ. The frozen macro executor retains its own internal 1 kJ return margin. This intervention therefore measures the final guard’s reserve, rather than a

consistent global reoptimization of every planning reserve. Additional diagnostics log action energy, time, key consumption, attempted and accepted frames, and pre-decision posterior mean and quantile exposure. Ground truth is used only when evaluating these recorded quantities.

17 Measurement basis and reproducibility mapping

Published drone key-generation experiments establish that actual measurements can constrain a deployment model. The inspected DroneKey author paper reports 2.4 GHz WiFi, about 140 transmitted packets per second, BCH(31,15), and a group-key rate of 89.5 bit/s for ten devices. The reciprocal probe delay, privacy mechanism, and group construction differ from this simulator. Those numbers provide context and do not calibrate the present functions. A publicly accessible UAV environmental-measurement record confirms field experiments but does not expose the parameter set needed for transfer. No claim of measurement-derived coherence, correlation threshold, reconciliation loss, or net key rate is made.

The detailed literature audit records primary URLs, access dates, and the boundary of each comparison. Additional source records belong to the supplementary provenance rather than extending the main article’s 15-reference list. The relevant sources are the DroneKey author manuscript and the UAV environmental measurement record. These records were accessed on September 19, 2026.

The base geometry, radio, PLKG, and energy functions are in `code/wes_orchestration/`. The V4 configuration and environment are in `evidence_v4_20260919/scripts/v4_config.py` and `evidence_v4_20260919/scripts/v4_env.py`. The stationary authentication process is inherited from `evidence_v2_20260731/scripts/v2_auth.py`. The common option catalogue, analytic forecast, feature order, and macro executor are in `evidence_v4_20260919/scripts/v4_policies.py`. The original gate is in `evidence_v5_20260919/scripts/v5_learning.py`. The revised gate and compact model are in `evidence_v6_20260920/scripts/v6_learning.py`. Frozen configuration records and source hashes accompany the numerical results.

The figure-packet trace starts with one initial snapshot. Each subsequent row is the state after an executed action, with separate fields for the pre-action position, energy, authentication cues, and requested action. The robustness NPZ traces use a different convention. Each row combines pre-action posterior, truth, and state values with increments from the executed interval. Those traces have no initial START row. Figures use one shared recorded episode rather than reconstructing a favourable schedule. True receiver overlays are evaluator-only. Figure selection and numerical outcome checks are retained with the figure packet. Desktop timing and model-file sizes describe the tested computer and do not establish airborne compute performance.

18 Reciprocity operating region and measurement context

This section examines the operating region of the reciprocity proxy in Section 10 and distinguishes a numerical consistency check from measurement calibration. The reciprocal measurement gap and the sampling interval are different quantities. The WARP measurements of Zhang and colleagues used a 0.06 ms data to acknowledgement gap and a 0.96 ms repeated transmission interval at 2.412 GHz. The experiments used indoor and laboratory environments. These settings support the distinction between the two timescales. The measurements do not calibrate the UAV proxy. The primary source is Experimental Study on Key Generation for Physical Layer Security in Wireless Communications, IEEE Access, volume 4, pages 4464 to 4477, 2016. The author manuscript was accessed on 20 September 2026.

For the implemented proxy, write $r_0 = 0.93$ and $L = \exp[-(\tau/T_c)^2]$. Positive effective yield requires

$$L > r_0, \quad \gamma > \frac{r_0}{L - r_0}. \quad (58)$$

The first condition follows from the upper bound $\rho < L$. The second follows by solving $\gamma L / (1 + \gamma) > r_0$. Consequently, even an arbitrarily high SNR requires

$$\frac{\tau}{T_c} < \sqrt{-\ln r_0} = 0.269389. \quad (59)$$

The assumed 5 ms gap therefore requires $T_c > 18.5605$ ms. Table 25 evaluates the three configured jitter values. The sampling cap of 100 Hz is inactive at these values. The effective sample intervals are the listed coherence times.

Table 25. Operating region of the assumed reciprocity proxy. The SNR bounds are strict. The limiting gross rate precedes the availability mask, privacy margin, and reconciliation retention.

Altitude m	T_c ms	f_s Hz	Minimum SNR dB	Limiting gross rate bit/s
60	38.462	26	12.422	316.410
100	33.333	30	12.895	327.437
140	27.778	36	13.873	313.667

The reproducibility script evaluates 126 combinations of altitude, measurement gap, and SNR by calling the frozen reciprocity and effective yield functions. Every point satisfies the derived condition. This numerical check establishes the operating region of the stated model. The check does not establish agreement with airborne measurements. The eight independent subbands, jitter to coherence mapping, correlation threshold, and reconciliation retention remain assumptions. The workspace contains no raw channel measurements that could calibrate these quantities. Key generation occupies a stationary service interval, so nominal cruise speed alone does not determine the assumed jitter coherence time.

18.1 Session key workload

A new 256 bit session key covers up to $B = 4$ Mbit, or 500 kB with decimal prefixes. This workload represents frequent key rotation across payload segments and makes finite fresh key inventories consequential. The workload is not an IEEE mandated rekeying period or a one time pad. Actual confidentiality also depends on the encryption mode and nonce management, which the simulator does not implement.

For a 400 Mbit task, ideal continuous packing requires $400/B$ keys. The $B = 1, 4$, and 16 Mbit conditions therefore require 102.4, 25.6, and 6.4 kbit of key material. Per transmission rounding can increase actual consumption. The load interventions quantify the dependence on this design assumption. They do not identify a recommended operational rekeying period or establish a cryptographic compromise bound.

19 Earlier comparisons and controlled ablations

This section preserves the additional comparisons referenced by the case-study guide in Section 3. The frozen records cover earlier fit variation, independent privacy and sensing controls, and cohort A exposure accounting. The analyses answer different questions from the fresh cohort B evaluation in Section 4.1. No historical result selects a V6 model.

19.1 Earlier objective values and fit variation

The earlier V4 cohort contains 1280 contexts from 320 scenario clusters and was already inspected before the V5 review. Its results remain useful for diagnosis but are separate from new confirmation. Values for learned families in the following table average the four fitted models within each identical context before interval estimation.

Table 26. Original fixed utility on the earlier eight map cohort. Brackets give 95 percent exploratory intervals.

Budget kJ	Policy	Utility and interval
Pooled	Fixed option	81.28 [72.13, 90.80]
Pooled	Sensing portfolio	89.55 [84.41, 95.49]
Pooled	Analytic	91.12 [86.27, 96.48]
Pooled	No sensing fixed	88.14 [82.26, 94.31]
Pooled	No sensing portfolio	97.73 [94.62, 100.40]
Pooled	Residual learner	95.03 [90.61, 99.77]
Pooled	No sensing learner	98.50 [95.21, 101.97]
Pooled	Fixed altitude learner	92.49 [87.57, 97.31]
Pooled	Fixed gateway learner	90.83 [85.37, 96.29]
70	Fixed option	57.75 [44.08, 73.41]
70	Sensing portfolio	57.75 [44.08, 73.41]
70	Analytic	57.80 [40.11, 77.60]
70	No sensing fixed	84.69 [74.87, 93.02]
70	No sensing portfolio	84.69 [74.87, 93.02]
70	Residual learner	69.89 [55.65, 84.82]
70	No sensing learner	82.74 [70.57, 93.90]
70	Fixed altitude learner	67.39 [55.87, 79.46]
70	Fixed gateway learner	70.67 [58.05, 84.43]
90	Fixed option	82.17 [71.61, 92.72]
90	Sensing portfolio	86.44 [79.17, 94.19]
90	Analytic	101.80 [99.46, 103.87]
90	No sensing fixed	89.29 [84.00, 94.81]
90	No sensing portfolio	91.73 [87.30, 95.61]
90	Residual learner	101.24 [98.33, 103.93]
90	No sensing learner	99.11 [94.90, 102.82]
90	Fixed altitude learner	97.82 [92.64, 102.08]
90	Fixed gateway learner	93.52 [88.30, 98.44]
110	Fixed option	90.97 [83.24, 98.23]
110	Sensing portfolio	107.01 [105.38, 108.23]
110	Analytic	102.47 [100.31, 104.39]
110	No sensing fixed	89.29 [84.08, 94.82]
110	No sensing portfolio	107.26 [105.49, 108.60]
110	Residual learner	104.30 [102.19, 106.09]
110	No sensing learner	106.11 [103.87, 107.77]
110	Fixed altitude learner	102.41 [100.12, 104.37]
110	Fixed gateway learner	99.33 [96.63, 101.71]
130	Fixed option	94.23 [87.99, 99.79]
130	Sensing portfolio	107.01 [105.38, 108.24]
130	Analytic	102.41 [100.25, 104.27]

Budget kJ	Policy	Utility and interval
130	No sensing fixed	89.29 [83.96, 94.72]
130	No sensing portfolio	107.26 [105.49, 108.60]
130	Residual learner	104.70 [102.83, 106.28]
130	No sensing learner	106.03 [104.06, 107.52]
130	Fixed altitude learner	102.33 [100.25, 104.17]
130	Fixed gateway learner	99.80 [97.54, 101.75]

Table 27. Separate fitted model means on the earlier cohort. Model fits share the underlying offline simulation data.

Policy	Fit	Utility	Secure %	Complete %	Energy kJ
Residual learner	0	93.89	90.91	87.42	78.58
Residual learner	1	96.40	93.02	91.17	75.62
Residual learner	2	94.53	91.24	89.45	77.76
Residual learner	3	95.32	91.84	89.92	79.45
No sensing learner	0	98.35	94.86	93.28	84.41
No sensing learner	1	98.85	95.59	93.28	87.42
No sensing learner	2	98.39	94.68	93.05	85.95
No sensing learner	3	98.40	94.98	93.05	83.22
Fixed altitude learner	0	92.70	90.68	89.22	72.99
Fixed altitude learner	1	92.65	90.84	89.22	72.98
Fixed altitude learner	2	92.52	90.81	89.45	72.27
Fixed altitude learner	3	92.07	90.08	89.22	74.62
Fixed gateway learner	0	90.94	91.15	88.52	71.72
Fixed gateway learner	1	89.00	89.78	86.88	72.39
Fixed gateway learner	2	91.63	91.64	89.84	72.61
Fixed gateway learner	3	91.76	91.94	90.31	72.86
Direct learner	0	96.06	92.68	90.55	78.13
Direct learner	1	95.73	92.38	90.31	77.01
Direct learner	2	96.46	92.52	91.02	78.68
Direct learner	3	96.12	92.51	91.02	79.65

The matched four fit comparison subtracts residual utility from direct utility. The mean difference is 1.06. Its interval extends from negative 0.14 to 2.63. This interval does not establish a residual specific advantage or equivalence.

19.2 Independent privacy and sensing crossings

The crossed experiment fixes gateway index one and altitude 60 m, as chosen in the earlier validation restrictions. It independently varies a fixed discard fraction of 0.65, posterior quantiles 0.95 and 0.99, and zero, two or four additional sensing dwells. Every cell pays the same two initial dwells. The 1440 executions use 80 earlier scene clusters at 70 and 90 kJ. No cross result selects a new controller.

The fixed 0.65 rule does not complete a mission at this selected gateway and altitude under these budgets. Configurations with zero delivery do not acquire a security advantage from having zero modeled exposure.

Table 28. Means for the fully crossed privacy and additional sensing settings. S denotes the additional dwell count and complete secure denotes completion adjusted modeled secure payload.

Budget kJ	Configuration	Utility	Complete %	Secure %	Exposed %	Energy kJ	Complete secure %
Pooled	Fixed 0.65 S0	-13.24	0.00	3.41	0.08	78.81	0.00

Budget kJ	Configuration	Utility	Complete %	Secure %	Exposed %	Energy kJ	Complete secure %
Pooled	Fixed 0.65 S2	-16.61	0.00	0.00	0.00	78.91	0.00
Pooled	Fixed 0.65 S4	-16.52	0.00	0.00	0.00	78.67	0.00
Pooled	Q95 S0	43.09	48.12	49.71	0.72	72.44	47.45
Pooled	Q95 S2	66.73	66.88	70.02	1.52	68.17	65.56
Pooled	Q95 S4	65.38	67.50	68.48	1.53	68.43	66.31
Pooled	Q99 S0	17.30	26.88	28.71	0.26	75.70	26.64
Pooled	Q99 S2	52.85	51.25	58.67	0.69	72.14	50.76
Pooled	Q99 S4	58.41	60.00	62.94	0.95	71.18	59.20
70	Fixed 0.65 S0	-15.12	0.00	0.00	0.00	68.75	0.00
70	Fixed 0.65 S2	-15.14	0.00	0.00	0.00	68.85	0.00
70	Fixed 0.65 S4	-15.15	0.00	0.00	0.00	68.95	0.00
70	Q95 S0	13.00	21.25	24.27	0.65	67.58	20.69
70	Q95 S2	52.85	53.75	57.96	0.94	65.13	53.06
70	Q95 S4	55.20	57.50	59.53	1.03	65.48	57.08
70	Q99 S0	5.39	16.25	17.66	0.13	68.15	16.17
70	Q99 S2	32.80	30.00	42.07	0.25	67.88	30.00
70	Q99 S4	42.58	43.75	49.57	0.51	67.65	43.51
90	Fixed 0.65 S0	-11.36	0.00	6.83	0.16	88.87	0.00
90	Fixed 0.65 S2	-18.09	0.00	0.00	0.00	88.98	0.00
90	Fixed 0.65 S4	-17.90	0.00	0.00	0.00	88.39	0.00
90	Q95 S0	73.18	75.00	75.15	0.79	77.29	74.21
90	Q95 S2	80.61	80.00	82.07	2.10	71.21	78.05
90	Q95 S4	75.57	77.50	77.44	2.03	71.38	75.55
90	Q99 S0	29.21	37.50	39.75	0.39	83.26	37.11
90	Q99 S2	72.90	72.50	75.27	1.13	76.40	71.52
90	Q99 S4	74.24	76.25	76.32	1.40	74.71	74.89

Table 29. All controlled pooled contrasts in the cross experiment. Each row changes privacy at a fixed sensing count or sensing at a fixed privacy rule.

Comparison	Utility difference	Complete secure difference pp	Exposed difference pp
Q95 S0 minus Fixed 0.65 S0	56.33 [37.18, 79.02]	47.45 [30.96, 67.30]	0.64 [-0.06, 2.10]
Q99 S0 minus Fixed 0.65 S0	30.54 [5.41, 60.40]	26.64 [5.48, 51.65]	0.18 [-0.15, 0.80]
Q99 S0 minus Q95 S0	-25.79 [-35.95, -15.92]	-20.81 [-29.05, -12.37]	-0.46 [-1.37, -0.01]
Q95 S2 minus Fixed 0.65 S2	83.34 [65.24, 100.52]	65.56 [49.85, 80.25]	1.52 [0.29, 3.34]
Q99 S2 minus Fixed 0.65 S2	69.46 [50.70, 89.64]	50.76 [33.78, 69.25]	0.69 [0.00, 2.02]
Q99 S2 minus Q95 S2	-13.88 [-21.99, -7.66]	-14.80 [-22.32, -7.89]	-0.83 [-1.70, -0.17]
Q95 S4 minus Fixed 0.65 S4	81.91 [63.41, 99.28]	66.31 [50.44, 81.03]	1.53 [0.19, 3.56]
Q99 S4 minus Fixed 0.65 S4	74.93 [56.56, 92.94]	59.20 [43.94, 74.92]	0.95 [0.00, 2.56]
Q99 S4 minus Q95 S4	-6.98 [-14.11, -2.21]	-7.11 [-15.51, -1.04]	-0.57 [-1.19, -0.13]
Fixed 0.65 S2 minus Fixed 0.65 S0	-3.37 [-5.36, -1.32]	0.00 [0.00, 0.00]	-0.08 [-0.20, 0.00]
Fixed 0.65 S4 minus Fixed 0.65 S0	-3.28 [-5.30, -1.24]	0.00 [0.00, 0.00]	-0.08 [-0.20, 0.00]
Fixed 0.65 S4 minus Fixed 0.65 S2	0.09 [0.02, 0.21]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Q95 S2 minus Q95 S0	23.64 [10.40, 37.42]	18.11 [6.53, 30.00]	0.80 [-0.01, 1.84]
Q95 S4 minus Q95 S0	22.29 [8.42, 36.89]	18.86 [6.98, 30.49]	0.81 [-0.19, 2.21]
Q95 S4 minus Q95 S2	-1.34 [-4.65, 2.75]	0.75 [-2.49, 5.73]	0.01 [-0.87, 0.80]
Q99 S2 minus Q99 S0	35.55 [16.36, 56.01]	24.12 [10.59, 39.51]	0.43 [-0.16, 1.34]
Q99 S4 minus Q99 S0	41.11 [20.34, 63.62]	32.56 [15.71, 50.17]	0.69 [-0.03, 1.89]
Q99 S4 minus Q99 S2	5.56 [1.43, 10.49]	8.44 [2.68, 15.45]	0.26 [-0.16, 1.07]

19.3 Exposure mapping sensitivity on fixed trajectories

The shadow accounting uses residual exposure raised to exponent a before retained material enters the shadow inventory. Exponents $a = 0.5, 1, 2$ retain every original observation, action, total inventory and energy value. This is an accounting sensitivity analysis on all 640 cohort A contexts and does not add independent executions or validate a cryptographic secrecy interpretation.

Table 30. Paired pooled utility differences under three residual exposure mappings.

Exponent	No sensing learner minus portfolio	Gate minus residual	Gate minus portfolio
0.50	4.81 [1.88, 7.79]	4.42 [0.33, 8.69]	3.30 [0.46, 6.29]
1.00	3.65 [1.59, 5.74]	4.11 [0.86, 7.43]	1.79 [−0.51, 4.03]
2.00	2.69 [1.36, 4.07]	4.17 [1.49, 6.82]	0.52 [−1.69, 2.30]

The pooled ordering of all eight policy means is unchanged across these three mappings. This limited stability does not establish robustness across other inference, physical model or workload changes. Both confirmation cohorts retain exponent one accounting. Detailed mapping values and all budget groups are preserved with the raw results.

20 Module timing and memory diagnostics

This section retains earlier measurements of individual modules and process memory. Section 4.5 gives the current complete software decision chain, and Section 4.6 reports training costs. The measurement boundaries differ, so module percentiles must not be summed to estimate chain latency. The desktop host has an Intel Core Ultra 9 275HX processor, 24 logical processors, approximately 32 GB of memory, and Windows 11. Python is version 3.13.9. Random forest prediction and numerical library pools use one thread. The measured workload contains 40 validation contexts from four maps, two budgets, and five missions per map and budget. The particle timing sample contains 128 observed updates along the resulting mandatory sensing trajectories.

Timing excludes scenario construction, model loading, and plotting. Static channel and candidate caches remain available, whereas forecasts of the current operational state are refreshed. Each reported component has its own measurement boundary. Forest prediction is already included in the complete option scoring measurement and must not be added again.

Component	Samples	Median ms	95th percentile ms	Maximum ms
Initial gate and forecasts	40	25.665	28.552	31.178
All 36 option scores and forecasts	40	29.116	33.029	37.174
Forest prediction alone	40	7.956	10.895	12.442
One service position replan	40	0.516	0.619	0.704
Particle update	128	0.639	2.246	3.923

Table 31. Measured desktop component latency. The 95th percentile is a sample latency quantile, not a confidence bound.

The serialized residual selector occupies 66,956,256 bytes, or 63.9 MiB. The serialized gate occupies 430,948 bytes, or 0.41 MiB. The uncompressed node and value arrays occupy 197,396,784 and 1,161,216 bytes, respectively. Process resident memory increases from 162,476,032 to 368,586,752 bytes after model loading. Peak process working set is 561,840,128 bytes, or approximately 536 MiB. These process measures include the interpreter, imported libraries, both models, and scenario arrays. Model file size therefore does not specify total memory demand.

The measured component times are below the four second simulation interval on this desktop. This comparison does not certify an airborne deadline because flight hardware, scheduling, input transfer, worst case workloads, and thermal limits have not been measured.

The energy model includes propulsion and an 18 W sensing increment during each SENSE action. HARVEST, TX, and AUTH actions incur a 5 W increment. General computation and idle sensor electronics are excluded. A learned choice can reduce modeled mission energy through service time or route changes, but the experiments do not establish a reduction in computing power.

The task energy efficiency is the ratio of summed completion adjusted modeled secure payload to summed mission energy

$$\eta = \frac{\sum_j 400(S_j/100)I_j}{\sum_j E_j} \text{ Mbit/kJ.} \quad (60)$$

The indicator I_j denotes strict completion of mission j . Failed missions remain in the energy denominator. This definition differs from averaging individual mission ratios. Paired bootstrap calculations recompute the ratio in each draw.

21 Earlier experiment records and file locations

The paths below locate earlier experiment records within the separate experiment archive described in Section 1.3. Section 1.4 indexes the current V6 records. The accompanying workspace retains the original manuscript and the frozen V4 evidence. The V5 directory contains the exact policy checkpoints, raw mission records, per-transition diagnostics, inference interventions, numerical summaries, and plotting programs. The file `evidence_v5_20260919/REPRODUCTION.md` gives the dependency chain, command sequence, and per-scenario access examples.

The cohort A confirmation records are `evidence_v5_20260919/data/confirmation_map*_budget*.npz`. Their paired analysis is `evidence_v5_20260919/data/results_confirmation_v5.json`. The independent audit is `evidence_v5_20260919/data/confirmation_independent_audit_v5.json`. Frozen file hashes are recorded in `evidence_v5_20260919/data/confirmation_freeze_v5.json`.

The fixed trajectory exposure transformations are recorded in `evidence_v5_20260919/data/results_mapping_confirmation_v5.json`. The privacy and sensing cross experiment is summarized in `evidence_v5_20260919/data/results_cross_v5.json`. The older cohort comparisons are summarized in `evidence_v5_20260919/data/results_learning_secondary_v5.json`.

The robustness archive uses `evidence_v5_20260919/data/robustness*_map*_budget*.npz`. The declared conditions and complete statistics are `evidence_v5_20260919/data/robustness_design.json` and `evidence_v5_20260919/data/robustness_statistics.json`. Computing measurements are stored in `evidence_v5_20260919/data/runtime_v5.json`. The earlier V5 illustrated trajectory and its replay checks are in `evidence_v5_20260919/data/figure_episode_v5.json`.

All figure curves and confidence intervals are computed from the saved records. Existing frozen files are retained when analyses and figures are regenerated. Source archives preserve relative paths so that the main article and this note can be compiled independently of unrelated older figure files.